

Final Project Description and Rubric

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Student (s) Signature: _____

Student Name(s):	Hessa Khalfan	Maryam Ali	Nourah Abdulla
Student HCT ID(s):	H00535166	H00535836	H00535834

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1.1 Process Identification and Description

1.1.1 Organization Name: Noon.com

Noon.com is a leading digital e-commerce platform launched in 2017, that has quickly established itself as a major player in the Middle East, primarily serving the UAE, Saudi Arabia, and Egypt. Founded with a massive initial investment of US\$1 billion from a consortium including Saudi Arabia's Public Investment Fund and Emirati businessman Mohamed Alabbar. Noon aimed to compete with global giants like Amazon (Miracuves, 2025). The company operates on a hybrid business model, combining a direct-to-consumer (D2C) division (where Noon acts as the seller) with a robust e-commerce marketplace that enables thousands of local and regional businesses to sell their products on the platform (The Story of Startups in the UAE, 2025). This dual approach allows Noon to offer a wide and diverse product catalog, including electronics, fashion, home goods, and groceries, while simultaneously fostering a dynamic ecosystem for small and medium-sized enterprises (SMEs).

Noon's strategic vision is based on its ambition to be a leader in "Arab First" e-commerce, specifically designed to meet the needs and preferences of consumers in the Middle East. This is reflected in its local marketing, customer support, and strong focus on building an integrated logistics and payment infrastructure. The platform's success rests on key pillars including competitive pricing, fast delivery services (such as same-day or next-day delivery), and a secure and integrated payment system (Noon Payments) which facilitates seamless transactions (Dcrayons, 2026).

1.1.2 Process Name: Customer Order Fulfillment (Directship Model)

The process selected for analysis is customer order fulfillment, which operates under Noon's **Direct Delivery** model. Also known as Delivery by Partner, this model is a core component of Noon's e-commerce strategy. It allows third-party sellers to manage their inventory and fulfill orders directly from their warehouses, utilizing Noon's platform and logistics network for final delivery to the customer (Noon Partners, n.d.-a).

This process was chosen due to its pivotal role in customer experience. Unlike Noon's FBN (Fulfilled by Partner) model, where Noon handles all warehousing and delivery. The Direct Delivery model involves complex interactions between several independent parties: the customer, Noon's Order Management System (OMS), the third-party seller, the payment gateway, and the logistics carrier. The efficiency and reliability of this process directly impact customer satisfaction, seller performance, and Noon's overall brand reputation, making it a rich and important subject for business process analysis.

1.1.3 Process Description

The direct delivery order fulfillment process is a series of coordinated activities that begins the moment a customer places an order and ends with its successful delivery. The following outlines the key steps:

1. Order Placement and Process Start: The process begins when a customer, using the Noon mobile app or website, selects products sold by a third-party direct delivery partner and

completes the checkout process. This includes confirming the products, providing the delivery address, and selecting a payment method.

2. Payment Approval: Upon order placement, Noon's Central Order Management System (OMS) immediately initiates a transaction request. It communicates with an external payment gateway (such as Noon Payments, which in turn connects to Visa, Mastercard, or other financial institutions) to approve the customer's payment. If the payment is approved, the seller will receive the order.

3. Merchant Notification and Order Acceptance: Once payment is approved, the OMS automatically sends a notification to the relevant seller. The seller receives this new order. The order appears in the "Pending" tab, awaiting seller action.

4. Order Preparation, Packaging, and Airway Billing: The seller is now responsible for the actual preparation of the order. First, they must select the correct products from their inventory. Next, they must package the products according to Noon's strict packaging guidelines, often using official Noon-branded packaging materials to ensure consistent customer experience. A key step at this stage is issuing a unique shipping label (Shipment details). This is typically done by scanning a pre-printed QR code on the Noon packaging, which links the physical package to the order in the system (Noon Partners, n.d. - A).

5. Logistics Coordination and Pickup Request: After the package is prepared, the seller submits a formal pickup request. This action signals to Noon's system that the shipment is ready for pickup. The system then coordinates with its logistics arm, Noon Logistics, or with an integrated third-party carrier (such as Aramex) to assign a driver to the seller's warehouse location.

6. Initial Delivery and Status Update: The assigned driver arrives at the seller's location and scans shipping label (Shipment details) on the package. This scan is a critical event, as it updates the order status to "Shipped" within the Noon system. This status change is immediately visible to the customer via their order tracking page. The driver then transports the package to Noon's regional sorting and forwarding center.

7. Final Delivery and Proof of Delivery: From the Noon center, the final delivery driver delivers the package to the customer's specified address. Upon successful delivery, the driver is required to obtain proof of delivery. This can be done by photographing the package at the customer's doorstep or, for high-value items, by entering a one-time password (OTP) sent to the customer's phone.

8. Transaction Completion and Seller Payment: The captured proof of delivery is sent to the Order Management System (Noon OMS), resulting in a final status update and classifying the order as "Delivered." This event officially concludes the order fulfillment process for the customer.

1.1.4 Integration with External and Internal Entities

The Direct Delivery model depends on smooth, real-time connections between Noon's main platform and different internal and external systems. These integrations are not just helpful additions. They are the foundation that makes the entire process possible.

External Entities:

- **Payment Gateways:** These are third party companies that handle online payments. When a customer places an order, Noon's Order Management System (OMS) securely sends the payment details to the gateway through an API call. The gateway then replies with either an approval or a rejection response in real time. Based on this response, Noon either confirms the order or cancels it. This real time integration is essential for keeping customer payment data safe and preventing fraud.
- **Third-Party Sellers:** Even though they are partners, sellers are separate businesses with their own stock and ways of working. The Seller Lab platform is the main integration point between Noon and its sellers. It allows two way data flow of information. Noon sends new order details to the seller through this system. In return, the seller sends status updates back to Noon, such as when a package is ready for pickup or when inventory levels change.
- **External Logistics Partners:** For some deliveries such as large items or cross-border shipping, Noon works with outside delivery companies such as Aramex or DHL. Noon's system integrates directly with these carriers' APIs. It sends them shipment details like shipment details and customer address. In return, Noon receives live tracking updates showing real time packages location, so customers can always see the status of their order from start to finish.

Internal systems:

- **Order Management System (OMS):** This is the central system that runs the entire process. It connects with all other systems, keeps the main record of each order's status, and manages the workflow from start to finish.
- **Noon Logistics Platform:** This is Noon's own system for managing its delivery drivers and warehouses. The OMS links with this platform to request pickups from sellers and to track packages during transit and final delivery to customers.
- **Customer Relationship Management (CRM) System:** Although it is not directly involved in the delivery process, the CRM system connects with the OMS. It records all customer interactions and order history. This information helps customer service agents solve problems that may come up, such as when a customer asks about a late delivery.

All these connections show how much technology is needed to make the Direct Delivery model work properly. A primary challenge in this model is data synchronization latency; for example, if a seller updates their stock in 'Seller Lab' but the integration with Noon's main platform lags, it could lead to 'ghost stock' where a customer buys an item that is sold out. If any of these systems fail to communicate correctly or experience delays, it can cause errors, canceled orders, and poor customer experience.

1.2 BPMN Process Model

Each participant is represented by a separate pool. Sequence flows (solid lines) show the progression of tasks within each participant's control, while message flows (dashed lines) represent the communication and data exchange between different participants.



Figure 1: The BPMN diagram

deliverable 1: Self-Reflection

deliverable 1- Self-Reflection: Nourah Abdulla Alghfeli - H00535834

Reflection on 1.1.1 Organization Name

I chose [Noon.com](https://noon.com) because it's a well-known regional e-commerce platform with a complex hybrid model. Researching its background helped me understand how company structure affects process design. The challenge was finding reliable internal details since companies don't publish everything. Next time, I would look for more official sources. I believe I clearly explained why Noon is suitable for process analysis.

Reflection on 1.1.3 Process Description

Writing about order placement and payment approval showed me how many steps happen before a seller even sees an order. The hardest part was balancing detail with clarity. I learned that good process descriptions need both accuracy and readability. If I improved this, I would add more technical details about how the OMS communicates with payment gateways.

Describing order preparation and delivery showed me how many handoffs occur in this model. Each handoff creates potential waste, connecting to our Week 6 waste analysis lessons. The challenge was explaining technical steps like airway billing in simple terms.

Reflection on Diagram Draft

Creating the first draft of the BPMN model was my most significant learning experience, as it involved translating our written description into a visual language for the first time. The initial challenge was twofold. First, I struggled with correctly representing the "conversation" between different participants using message flows, particularly how to show the interaction between the customer, the Noon system, and the payment gateway. Second, my initial attempt focused on getting the main "successful path" correct, but I found it difficult to visually represent the "unsuccessful paths," such as what happens if a payment fails.

This first draft, even with its imperfections, was a crucial step. It highlighted the need for clear decision points (Gateways) early in the process and sparked a key team discussion on how to handle exceptions. It taught me that a good process model isn't just a straight line but a map with multiple routes, and that the purpose of a first draft isn't to be perfect, but to start the conversation and uncover the real modelling challenges for the team to solve.

deliverable 1- Self-Reflection: Hessa Khalfan Al Ali- H00535166

Reflection on 1.1.2 Process Name & Rationale

My role was to ensure that the BPMN model accurately and logically represented the process. I chose the Direct Delivery model for our analysis, preferring it over Noon's simpler FBN (Fulfilled by Noon) model. I chose this

approach because the direct delivery process involves accurate interactions between multiple independent companies. I believed this path would provide a richer and more challenging theme for our project, allowing us to demonstrate a deeper understanding of business process integration and modeling.

Reflection on 1.2 BPMN Diagram and BPMN Diagram Description (Second Paragraph)

Following the initial draft, my core contribution was to implement the detailed logic of the process using the correct BPMN elements. I was responsible for defining the Gateways that control the process flow, especially the unhappy paths. For example, I designed logic for the (Delivery Successful?) Gateway, ensuring it had clear, mutually exclusive outcomes that either completed the process or initiated a re-delivery loop.

Moreover, I focused on the precise technical implementation of tasks and events. user-driven tasks and automated **Service Tasks** like Update Status to Shipped, and correctly modeling the communication between pools. Instead of just connecting tasks directly, I ensured we used the correct pattern, such as a Send Task triggering a Receive Event. This detailed work on gateways, task types, and message events was crucial for transforming the basic diagram into a technically accurate and professional model that reflects the real-world system interactions

deliverable 1- Self-Reflection: Maryam Ali Al Ali- H00535836

Reflection on 1.1.4 Integration

I chose Noon's Direct Delivery model because it perfectly illustrates the "hybrid" business operations we studied in class. In this model, Noon must work with outside partners that it does not directly control. I explained how Noon connects with payment gateways, sellers, and logistics partners using APIs and real-time data exchange. I also added challenges like data synchronization latency and "ghost stock" (when customers order items that are sold out). This shows I understand the risks of integration and how they affect the business.

Reflection on 1.2 BPMN Diagram Description

Writing the introduction required me to look at the whole diagram first. I needed to make sure the boundaries between internal and external systems were clear. From Week 1 lessons, I learned that process models help different stakeholders understand complex workflows. The challenge was keeping my description short but still informative. To improve next time, I would explain more clearly why we chose these five specific participants for the Direct Delivery process.

Reflection on Final Diagram Check

My main job was checking that we used Pools and Message Flows correctly to show independent organizations. Using the Bizagi Modeler tool helped me maintain professional standards. The tool makes it easy to see the difference between sequence flows (solid lines inside each pool) and message flows (dashed lines between pools). I made sure all communication between different pools used message flows, while activities inside each pool used sequence flows. This final check helped make sure our diagram follows BPMN standards and accurately represents the real-world order fulfillment process.

2.1 Qualitative Analysis

2.1.1 Value-Added Analysis

In the Noon.com Direct Delivery order fulfillment process, each task has been classified into Value-Adding (VA), Business Value-Adding (BVA), or Non-Value-Adding (NVA) activities. This classification helps in identifying inefficiencies and streamlining the process.

Task Classification

Task	Classification	Justification
Complete checkout	VA	Directly fulfills the customer's goal of placing an order.
Authorized payment transaction	BVA	Essential for the business to secure funds and prevent fraud.
Cancel order	BVA	A necessary business control mechanism that protects Noon from processing and shipping unpaid orders, thereby preventing financial loss.
Process order	BVA	Necessary for the business to handle the order.
Notify Seller	BVA	Necessary communication for the business to fulfill the order.
Pick product from inventory	VA	Directly contributes to getting the physical product to the customer.
Package order	VA	Ensures the product is safely prepared for the customer.
Issue shipment details	BVA	Required for logistics tracking and management.
Submit pick up request	NVA	A manual coordination step that causes waiting time; should be automated.
Assign driver	BVA	Necessary for logistics coordination.
Scan Shipment details	BVA	Important for internal tracking and accountability.
Update status to shipped	BVA	Keeps the system updated; provides visibility.
Deliver to noon's warehouse	NVA	Pure transportation and waiting step; doesn't add direct value to the product itself.

Deliver to customer	VA	The core value proposition: getting the product to the customer.
Schedule Re-delivery attempt	NVA	Rework caused by a failed initial delivery attempt.
Return shipment to origin	NVA	Pure waste/rework due to ultimate delivery failure.
Capture Proof of Delivery (POD)	BVA	Protects the business from liability and confirms completion.
Update Status to Delivered	BVA	Closes the loop in the system for accounting and customer service.

Evaluation and Recommendations for Eliminating NVA Activities: The analysis reveals that tasks such as "submit pickup request", "deliver to noon's warehouse", and "Schedule Re-delivery attempt" are Non-Value-Adding (NVA). While physical transportation cannot be entirely eliminated in e-commerce, the waiting and coordination aspects can be minimized.

- **Recommendation 1 (Automate submit pickup request):** We recommend that Noon integrates its Order Management System (OMS) directly with the Seller platform to fully automate the pickup request step. Once the seller enters the shipment details, the system automatically generates a pickup request and sends it to the designated shipping company without any manual intervention.
- **Recommendation 2 (Reduce deliver to noon's warehouse):** For retailers with high sales volumes in specific geographic areas, Noon should consider a direct customer distribution center model. Instead of going to a central warehouse, drivers would collect packages from multiple local retailers and deliver them to a smaller, local sorting point for immediate distribution to customers. This would significantly reduce transportation costs by eliminating a key stage in the delivery journey.
- **Recommendation 3 (Eliminate Schedule Re-delivery attempt):** We recommend that Noon implement a network of smart lockers and pickup points across the UAE. If the first delivery attempt fails, the order management system should automatically send the customer a notification including the nearest available locker location and a secure pickup code. This transforms the NVA rework of re-delivery into a VA step (Provide Pickup Option) from the customer's perspective, giving them control and ensuring successful delivery.
- **Recommendation 4 (Return shipment to origin):** We recommend that Noon implement a policy to improve reverse logistics to reduce waste from returned undelivered shipments. After three failed delivery attempts, Noon should implement a smart sorting system. Undamaged products are returned to the platform as "open box" offers, while damaged products are disposed of locally or donated. This approach tackles the NVA rework of reverse logistics by creating a BVA process (reselling as open box) which recovers revenue and reduces waste. This creates a

new Business Value-Adding (BVA) process that recovers revenue, reduces shipping costs, and minimizes environmental waste.

2.1.2 Waste Analysis

Based on the Lean principles, we identified the following types of waste in the current process:

- 1 **Transportation (Move):** Moving the package from the Seller to Noon's warehouse, and then to the customer, involves double handling. Recommendation: Enable direct shipping from Seller to Customer for specific regions.
- 2 **Waiting (Hold):** The Seller must wait for the Logistics Carrier to arrive after submitting the pickup request. The package also waits at the warehouse for sorting. Recommendation: Optimize driver routing algorithms to reduce pickup wait times.
- 3 **Overprocessing (Overdo):** Scanning the shipment details multiple times (at pickup, at warehouse arrival, at warehouse departure). Recommendation: Use RFID tags or advanced barcode systems that allow bulk scanning without manual work.
- 4 **Defects/Rework (Overdo):** Failed deliveries leading to re-delivery attempts or returns. Recommendation: Improve address verification during checkout and provide drivers with better GPS navigation tools.

2.2 Quantitative Analysis (Time)

To analyze the time efficiency of the process, we assigned realistic time values to each task.

Table 1: Process Task Time Analysis

Task	Processing Time (PT)	Waiting Time (WT)	Cycle Time (CT = PT + WT)	Resources
Complete checkout	2 mins	0 mins	2 mins	1 (System)
process order	1 min	1 min	2 mins	1 (System)
Authorized payment transaction	1 min	2 mins	3 mins	1 (System)
Notify Seller	1 min	0 mins	1 min	1 (System)
Pick product from inventory	15 mins	30 mins	45 mins	2 (Warehouse Staff)
package order	10 mins	5 mins	15 mins	2 (Packers)
issue shipment details	2 mins	0 mins	2 mins	1 (System)
submit pickup request	1 min	0 mins	1 min	1 (System)

assign driver	2 mins	10 mins	12 mins	1 (System)
scan shipment details	2 mins	0 mins	2 mins	1 (Driver)
Update status to shipped	1 min	0 mins	1 min	1 (System)
deliver to noon's warehouse	30 mins	60 mins	90 mins	1 (Driver)
deliver to customer	25 mins	120 mins	145 mins	1 (Driver)
Capture Proof of Delivery (POD)	2 mins	0 mins	2 mins	1 (Driver)
Update Status to Delivered	1 min	0 mins	1 min	1 (System)

Calculations (Happy Path):

- **Value Added Time (Total PT):** $2 + 1 + 1 + 1 + 15 + 10 + 2 + 1 + 2 + 2 + 1 + 30 + 25 + 2 + 1 = 96$ minutes
- **Total Cycle Time (Total CT):** $2 + 2 + 3 + 1 + 45 + 15 + 2 + 1 + 12 + 2 + 1 + 90 + 145 + 2 + 1 = 324$ minutes
- **Cycle Time Efficiency (CTE):** $(\text{Theoretical Cycle Time} / \text{Total Cycle Time}) * 100 = (96 / 324) * 100 = 29.6\%$

Interpretation and Bottlenecks: A CTE of 29.6% indicates that the process is heavily burdened by waiting times, which is typical for logistics operations. The primary bottlenecks are:

- **deliver to customer (CT = 145 mins):** High waiting time due to traffic, routing, and waiting for the customer to receive the package.
- **deliver to Noon's warehouse (CT = 90 mins):** Significant transportation and waiting time for sorting.
- **Pick product from inventory (CT = 45 mins):** Manual searching in large seller warehouses causes delays.

2.3 Quantitative Analysis (Cost)

We calculated the labor cost based on the human resources involved in the process. We assume an average daily volume of 500 orders.

Table 2: Labor Cost Analysis per Order

Actor / Resource	Hourly Rate (AED)	PT per Order (mins)	Cost per Order (AED)
Seller Warehouse Staff	20 AED	15 mins	$(20/60) * 15 = 5.00$ AED

Actor / Resource	Hourly Rate (AED)	PT per Order (mins)	Cost per Order (AED)
Seller Packer	18 AED	10 mins	$(18/60) * 10 = 3.00$ AED
Logistics Driver	25 AED	59 mins (2+30+25+2)	$(25/60) * 59 = 24.58$ AED
Total Labor Cost per Order			32.58 AED

Full-Time Equivalent (FTE) Calculation: Assuming an 8-hour workday (480 minutes) and 500 orders per day:

Formula: $(\text{Average volume of orders} * \text{PT per min}) / \text{min per workday}$

- **Warehouse Staff FTE:** $(500 \text{ orders} * 15 \text{ mins}) / 480 \text{ mins} = 15.6$ (approx. 16 employees)
- **Packer FTE:** $(500 \text{ orders} * 10 \text{ mins}) / 480 \text{ mins} = 10.4$ (approx. 11 employees)
- **Driver FTE:** $(500 \text{ orders} * 59 \text{ mins}) / 480 \text{ mins} = 61.4$ (approx. 62 employees)
- **Total FTE Required:** $16 + 11 + 62 = 89$ Employees

Interpretation: The total labor cost per order is **32.58 AED**. The logistics driver accounts for the largest portion of the cost (24.58 AED) and requires the highest number of FTEs (62 drivers). Optimizing driver routes and reducing delivery times will have the most significant impact on reducing overall process costs.

2.4 Quantitative Analysis (Process Variability)

Metric Selected and Data Source Justification

To monitor the stability of the fulfillment process, we tracked the *Number of Orders Fulfilled (in thousands)* over a 24-month period. This metric was chosen because order volume directly impacts resource allocation, warehouse capacity, and logistics planning. Since Noon.com's internal operational data is proprietary and not publicly available, the dataset used for this analysis is simulated. It is modeled on typical e-commerce growth patterns and seasonal fluctuations observed in the Middle Eastern market, assuming a baseline of approximately 50,000 to 55,000 orders per month with periodic seasonal spikes (e.g., during "Yellow Friday" sales or Ramadan).

Table 3: Data: Number of Orders Fulfilled (in thousands) over 24 months.

M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
52	50	53	59	51	51	60	56	51	56	52	52
M13	M14	M15	M16	M17	M18	M19	M20	M21	M22	M23	M24
56	46	47	54	52	51	59	54	51	66	58	60

Step 1: Average Trade (Orders)

AVG(X) =

$$(52+50+53+59+51+51+60+56+51+56+52+52+56+46+47+54+52+51+59+54+51+66+58+60) / 24 = 54.04$$

Step 2: Monthly Moving Ranges

Table 4: Monthly Moving Ranges

Month	Calculation	MR
M2	50 - 52	2
M3	53 - 50	3
M4	59 - 53	6
M5	51 - 59	8
M6	51 - 51	0
M7	60 - 51	9
M8	56 - 60	4
M9	51 - 56	5
M10	56 - 51	5
M11	52 - 56	4
M12	52 - 52	0
M13	56 - 52	4
M14	46 - 56	10
M15	47 - 46	1
M16	54 - 47	7
M17	52 - 54	2
M18	51 - 52	1
M19	59 - 51	8
M20	54 - 59	5
M21	51 - 54	3
M22	66 - 51	15
M23	58 - 66	8
M24	60 - 58	2

Step 3: Average Moving Range

$$\text{AVG(MR)} = (2+3+6+8+0+9+4+5+5+4+0+4+10+1+7+2+1+8+5+3+15+8+2) / 23 = 4.87$$

Step 4: UCL and LCL for the Moving Range Chart

Using constants D1 = 0 and D2 = 3.267

$$\text{UCL} = \text{AVG}(\text{MR}) \times D2 = 4.87 \times 3.267 = 15.91$$

$$\text{LCL} = \text{AVG}(\text{MR}) \times D1 = 4.87 \times 0 = 0$$

Step 5: Natural Process Limits (UNPL & LNPL)

Using constant 2.66

$$\text{UNPL} = \text{AVG}(\bar{X}) + (2.66 \times \text{AVG}(\text{MR})) = 54.04 + (2.66 \times 4.87) = 54.04 + 12.95 = 66.99$$

$$\text{LNPL} = \text{AVG}(\bar{X}) - (2.66 \times \text{AVG}(\text{MR})) = 54.04 - (2.66 \times 4.87) = 54.04 - 12.95 = 41.09$$

Table 5: Summary Statistics

Statistic	Value (in thousands of orders)
Mean (X)	54.04
Upper Natural Process Limit (UNPL)	66.99
Lower Natural Process Limit (LNPL)	41.09
Mean Moving Range (MR)	4.87
Upper Control Limit for MR (UCL_MR)	15.91
Lower Control Limit for MR (LCL_MR)	0.00

Process Control Chart (I-MR) Based on the 24-month simulated dataset, we generated an Individual values and Moving Range (I-MR) process control chart to analyze process variability.

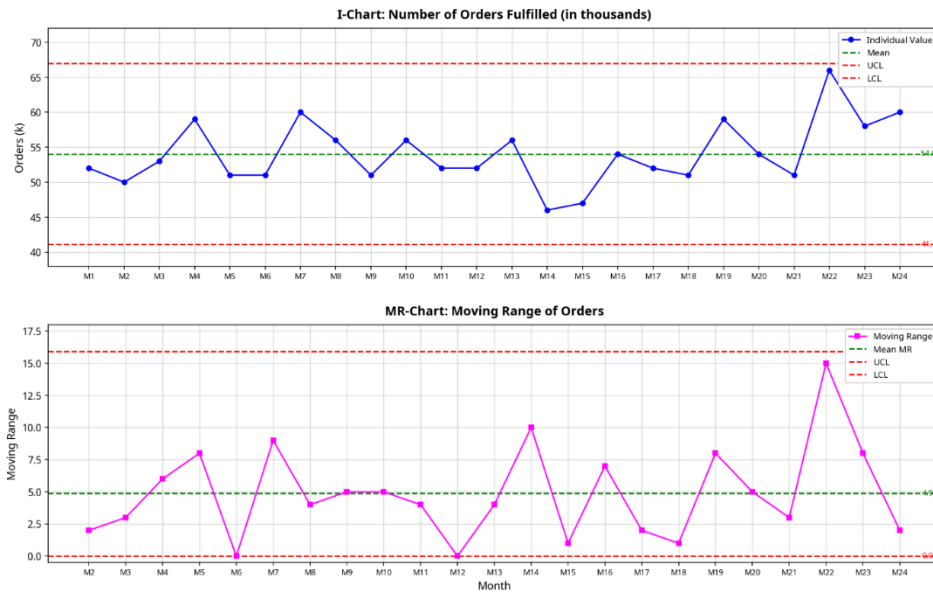


Figure 2: control_chart_noon

Interpretation and Business Context

- I-Chart (Individuals):** The top chart tracks the actual number of orders fulfilled each month. All 24 data points fall strictly within the Upper Control Limit (66.99k) and Lower Control Limit (41.09k). The data points fluctuate naturally around the mean (54.04k) without any consecutive runs of 7 or more points on one side of the mean, indicating that the process volume is statistically stable and subject only to common cause variation.
- MR-Chart (Moving Range):** The bottom chart tracks the absolute difference in order volume between sequential months, measuring the volatility of the process. All moving ranges are well within the control limits (below 15.91k), confirming that the month-to-month variation is predictable and stable.
- Business Implications and Seasonal Variation:** While the process is stable, Month 22 shows a significant spike to 66,000 orders, approaching the UCL. In a real-world context for Noon.com, this likely represents a predictable seasonal event, such as the annual "Yellow Friday" sale in November. Because this spike did not breach the UCL, the system successfully absorbed the shock without breaking down. This stability means Noon's Direct Delivery fulfillment process is highly predictable, allowing management to forecast capacity, hire temporary staff, and allocate logistics resources reliably.
- Conclusion:** The process is currently in a state of statistical control. There are no special causes of variation that require immediate corrective action. Management should continue monitoring the process; however, if the volume consistently approaches or exceeds the UCL in future months due to sustained business growth, the control limits will need to be recalculated to establish a new baseline.

Deliverable 2: Self-Reflection

Collaborative Effort (Section 2.1.1: Value-Added Analysis)

Before dividing the individual tasks, our team worked collaboratively on Section 2.1.1 (Value-Added Analysis). We discussed and classified each of the 20 tasks as VA, BVA, or NVA. One of our key discussions was about the "Cancel order" task. We debated whether it should be NVA, since it only happens when payment fails. However, after analyzing the business context, we agreed to keep it as BVA, because it is a necessary step that protects Noon from shipping unpaid orders and prevents financial loss. This discussion helped us better understand how to apply BPM concepts to real business situations.

deliverable 2- Self-Reflection: Maryam Ali Al Ali- H00535836

Assigned Sections: 2.2 Quantitative Analysis (Time), Quantitative Analysis Calculations (Control Chart Math), and Summary Statistics Table.

Reflection: My contribution focused mostly on the mathematical and quantitative aspects of Deliverable 2. I calculated the Cycle Time (CT), Theoretical Cycle Time (TCT), and Cycle Time Efficiency (CTE) for the entire process. The most challenging part was assigning realistic time values to each of the 20 tasks. By doing so, I discovered that the CTE was only 29.6%. It mathematically proved that the majority of the process time is spent waiting (e.g., waiting for a driver or waiting in the warehouse), and it clearly identified "deliver to customer" as the primary bottleneck.

Furthermore, I performed the manual step by step statistical calculations for the I-MR control chart. I calculated the mean (54.04k), the 23 monthly moving ranges, the average moving range (4.87), and the Natural Process Limits (UNPL = 66.99, LNPL = 41.09). The challenge was ensuring absolute precision in these calculations, so they perfectly matched the python generated chart. This exercise strengthened my understanding of how theoretical formulas translate into visual process monitoring. In future projects, I would improve the cost analysis by including other costs beyond labor, such as packaging, fuel, and warehouse expenses. I would also adjust the FTE calculations based on seasonal demand, like increasing drivers during "Yellow Friday," instead of using a fixed number of 500 orders per day.

deliverable 2- Self-Reflection: Hessa Khalfan Al Ali- H00535166

Assigned Sections: Evaluation and Recommendations for Eliminating NVA Activities, Metric Selected and Data Source Justification, and Writing the Python Code for Control Chart.

My primary responsibility was to develop realistic recommendations for eliminating or reducing non-value-added activities. When evaluating NVA tasks like delivering to noon's warehouse and return shipment to origin, my initial thought was to simply suggest eliminating them. But I soon realized that physical transportation and customs clearance are essential to a real-world cross-border e-commerce model like Noon's. The challenge lies in finding practical, not theoretical, solutions. I overcame this challenge by studying Noon's

actual operations, which led me to recommend advanced solutions, such as using the KEZAD automated distribution center to expedite the sorting process and implementing an improved reverse logistics policy (like selling returned products as open box offers) instead of costly international returns.

I was also responsible for the quantitative aspect of our analysis. I explained the Metric Selection and Data Source section. Because Noon's internal performance data is not publicly available, I created a 24-month simulated dataset for our control chart. This dataset was based on real-world observations of the e-commerce market to ensure that the control chart analysis was realistic and meaningful.

In addition, I wrote Python code to create an I-MR monitoring chart. The challenge here was ensuring the code accurately reflected the statistical formulas explained in the lecture (using the constants $D3=0$ and $D4=3.267$). Seeing the chart's visual output helped me understand how to monitor process stability. In future projects, I would refine my analysis by incorporating more predictive AI models into the recommendations to further reduce waiting times before they even occur.

deliverable 2- Self-Reflection: Nourah Abdulla Alghfeli - H00535834

Assigned Sections: 2.1.2 Waste Analysis, 2.3 Quantitative Analysis (Cost), and Detailed Interpretation and Business Context of the Control Chart.

Reflection: I was assigned the task of identifying the types of waste in our process, calculating the cost of labor, and explaining the business context of the control chart. The most difficult task was identifying the waste in our process. The e-commerce business has minimal waste because it is a highly digital business. However, I was able to successfully identify the Waiting, Transportation, Overprocessing, and Inventory wastes by mapping the physical flow of goods. Calculating the cost of labor was the second most difficult task. The difficulty was in estimating the salary of the employee. I was able to successfully complete this task. I learned that the logistics driver role consumes the majority of the cost of the labor. The cost of the logistics driver role is 24.58 AED out of 32.58 AED. Therefore, this is the area where Noon should focus in order to cut costs.

Interpreting the control chart was the most insightful task. I was required to explain the business context of the control chart. I was required to explain why the sudden spike in the number of orders in Month 22, which was 66,000 orders, does not mean the process was out of control. I was able to explain this statistical data in the context of the real business world. The business event was Noon's "Yellow Friday" sale. I was able to explain this business event in the context of the statistical data. Therefore, the system was able to handle this business event without going above the Upper Natural Process Limit. The business would not be able to handle this business event if it were above the Upper Natural

Process Limit. Therefore, the system was in control. If I were to redo this task in the future, I would be more successful in the cost evaluation section.

3. Service Oriented Architecture

To support the integration between the independent pools (Customer, OMS, Payment Gateway, Seller, Logistics Carrier) identified in our BPMN model, we have designed Service-Oriented Architecture (SOA) using RESTful web services.

Below are the API specifications for all 20 tasks in our process model, mapping the process tasks to web services using JSON data formats. Each service includes a rationale explaining why the specific RESTful VERB was chosen and how it aligns with the business process, ensuring a logical and well-justified architecture.

1. Select Items on App

Rationale: This task involves the customer browsing and retrieving product information from the catalog.

The [GET](#) method is used because it is a safe, read-only operation that fetches data without modifying the server state.

Field	Value
Method/VERB:	GET
URI:	www.noon.com/resources/Products
JSON Example:	<pre>{ "productId": "SKU-IPHONE15", "name": "iPhone 15", "price": 345.50, "stockStatus": "IN_STOCK" }</pre>

2. Complete Checkout

Rationale: The customer submits their cart and payment details to create a new order. The [POST](#) method is used because it creates a new resource (an order) on the server.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Orders
JSON Example:	<pre>{ "customerId": "CUST-102938", </pre>

	<pre>"cartId":"CART-5566", "totalAmount":345.50, "shippingAddress":"Dubai, UAE" }</pre>
--	---

3. Process Order

Rationale: The OMS receives the checkout request and initiates the internal order processing workflow. [POST](#) is used to trigger this backend processing event.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Orders
JSON Example:	<pre>{ "orderId":"ORD-987654321", "status":"PROCESSING_STARTED", "timestamp":"2026-03-27T10:05:00Z" }</pre>

4. Authorize Payment Transaction

Rationale: The OMS sends a request to the external Payment Gateway to authorize funds. [POST](#) is used because it securely transmits sensitive payment details to initiate a transaction.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Transactions
JSON Example:	<pre>{ "orderId":"ORD-987654321", "amount":345.50, "currency":"AED", "paymentMethod":{"type":"CREDIT_CARD", "token":"tok_visa_123"} }</pre>

5. Cancel Order

Rationale: If the payment is declined, the OMS must update the order status to canceled.
[PUT](#) is used because it updates an existing resource (the order) with a new status.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Orders/ORD-987654321
JSON Example:	<pre>{ "orderId": "ORD-987654321", "cancelReason": "PAYMENT_DECLINED", "status": "CANCELED" }</pre>

6. Notify Seller

Rationale: Once payment is approved, the OMS sends a notification to the Seller's system.
[POST](#) is used to push this new notification event to the seller.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Notifications
JSON Example:	<pre>{ "orderId": "ORD-987654321", "sellerId": "SEL-554433", "items": [{ "sku": "SKU-IPHONE15", "quantity": 1 }] }</pre>

7. Receive Order

Rationale: The seller's system fetches the details of the newly assigned order to begin processing. [GET](#) is used to retrieve the specific order details from the OMS.

OMS:

```
[
  {"orderId": "ORD-987654321", "customerName": "Ahmed Ali", "status":
"PENDING_SELLER_ACTION"},
  {"orderId": "ORD-111222333", "customerName": "Sara Mohammed", "status":
"PENDING_SELLER_ACTION"}
]
```

Field	Value
Method/VERB:	GET
URI:	www.noon.com/resources/Orders/ORD-987654321
JSON Request Example:	[{ "orderId": "ORD-987654321" }]

8. Pick Product from Inventory

Rationale: The seller updates their local inventory system to reflect that the item has been picked. [PUT](#) is used to update the existing inventory count.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Inventory/SKU-IPHONE15
JSON Answer Example:	{ "sku": "SKU-IPHONE15", "quantityPicked": 1, "remainingStock": 49, "pickerId": "EMP-001" }

9. Package Order

Rationale: The seller updates the order status in their system to indicate packaging is complete. [PUT](#) is used to modify the state of the existing order.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Orders/ORD-987654321
JSON Example:	<pre>{ "orderId":"ORD-987654321", "internalStatus":"PACKAGED", "packageWeightKg":0.5 }</pre>

10. Issue Shipment Details

Rationale: The seller generates the shipping label and tracking information. [POST](#) is used because it creates a new shipment record associated with the order.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Shipments
JSON Example:	<pre>{ "orderId":"ORD-987654321", "trackingNumber":"TRK-112233", "labelUrl": "https://seller.com/labels/112233.pdf" }</pre>

11. Submit Pickup Request

Rationale: The Seller notifies the OMS that the package is ready, requesting a driver. [POST](#) is used to create a new pickup request in the OMS.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/PickupRequests
JSON Example:	<pre>{ "sellerId": "SEL-554433", "orderId": "ORD-987654321", "warehouseId": "WH-DXB-01", "readyTime": "2026-03-27T14:00:00Z" }</pre>

12. Receive Pickup Request

Rationale: The OMS retrieves the list of pending pickup requests to process them.

[GET](#) is used to fetch this list without altering data.

Field	Value
Method/VERB:	GET
URI:	www.noon.com/resources/PickupRequests
JSON Example:	<pre>[{ "pickupRequestId": "PR-778899", "orderId": "ORD-987654321", "location": "WH-DXB-01" }]</pre>

13. Assign Driver

Rationale: The OMS assigns a specific logistics driver to the pickup request.

[PUT](#) is used because it updates the existing pickup request with the driver's ID.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/PickupRequests/PR-778899
JSON Example:	<pre>{ "pickupRequestId": "PR-778899", "driverId": "DRV-9988", "carrierId": "CAR-ARAMEX" }</pre>

14. Receive Pickup Task

Rationale: The driver's mobile app fetches their assigned tasks for the day. [GET](#) is used to retrieve the task list from the logistics server.

Field	Value
Method/VERB:	GET
URI:	www.noon.com/resources/Tasks
JSON Example:	<pre>[{ "taskId": "TSK-554", "type": "PICKUP", "location": "WH-DXB-01", "orderId": "ORD-987654321"}, { "taskId": "TSK-555", "type": "DELIVERY", "location": "WH-DXB-02", "orderId": "ORD-111222333"}]</pre>

15. Scan Shipment Detail

Rationale: The driver scans the barcode upon picking up the package, sending the event to the server. [POST](#) is used to record this new scan event.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Scans
JSON Example:	<pre>{ "trackingNumber": "TRK-112233", "scanType": "PICKED_UP", "driverId": "DRV-9988", "timestamp": "2026-03-27T15:30:00Z" }</pre>

16. Update Status to Shipped

Rationale: The Logistics Carrier sends the scan event to the OMS to update the customer-facing status. [PUT](#) is used to modify the order's status to "Shipped".

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Orders/ORD-987654321
JSON Example:	<pre>{ "status": "SHIPPED", "trackingNumber": "TRK-112233", "carrierId": "CAR-ARAMEX" }</pre>

17. Deliver to Noon Warehouse

Rationale: If the package goes through a sorting facility, the driver updates the location. [PUT](#) is used to update the current location of the shipment.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Shipments/TRK-112233
JSON Example:	<pre>{ "trackingNumber": "TRK-112233", "currentLocation": "SORTING_HUB_DXB", "status": "IN_TRANSIT" }</pre>

18. Deliver to Customer

Rationale: The driver arrives at the customer's location and initiates the handover process. [POST](#) is used to log the delivery attempt event.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/Deliveries
JSON Example:	<pre>{ "trackingNumber": "TRK-112233", "driverId": "DRV-9988", "gpsCoordinates": "25.2048, 55.2708" }</pre>

19. Capture Proof of Delivery

Rationale: The driver uploads the customer's signature or photo as proof. [POST](#) is used to create and store this new proof document.

Field	Value
Method/VERB:	POST
URI:	www.noon.com/resources/ProofOfDelivery
JSON Example:	<pre>{ "trackingNumber": "TRK-112233", "signatureImage": "base64_encoded_string", "recipientName": "Ahmed Ali" }</pre>

20. Update Status to Delivered

Rationale: The Logistics Carrier notifies the OMS that the delivery is complete. [PUT](#) is used to finalize the order status to *Delivered*.

Field	Value
Method/VERB:	PUT
URI:	www.noon.com/resources/Orders/ORD-987654321
JSON Example:	<pre>{ "status": "DELIVERED", "deliveryTime": "2026-03-28T11:45:00Z", "podReference": "POD-998877" }</pre>

3.1 Individual Self-Reflections (Deliverable 3)

deliverable 3- Self-Reflection: Nourah Abdulla Alghfeli - H00535834

For Deliverable 3, I was responsible for mapping tasks such as Process Order, Cancel Order, Package Order, Issue Shipment Details, Scan Shipment Detail, and Deliver to Noon Warehouse. The process of identifying these services from our BPMN diagram was straightforward because these tasks represent clear actions or state changes.

The RESTful API specification helped me define the architecture by forcing me to think about which HTTP VERB matches the action. For example, I used **PUT for Cancel Order** because it updates an existing order's status, rather than creating a new one. Similarly, for **Process Order**, I utilized the **POST** HTTP verb. This was chosen because processing a new order typically involves creating a new resource or submitting data to the server to initiate a new process, which aligns with the semantics of a POST request.

One challenge was ensuring the JSON data accurately reflected the specific details needed for each task, such as the cancellation reason for a Cancel Order request or the warehouse location pertinent to Deliver to Noon Warehouse.

In the future, I could improve my analysis by considering how these APIs handle errors or exceptions, making the architecture even more robust. Specifically, I would design retry mechanisms for failed payment authorization calls, which would enhance the system's resilience and ensure smoother operation even when external dependencies encounter temporary issues.

deliverable 3- Self-Reflection: Maryam Ali Al Ali- H00535836

My contribution to Deliverable 3 focused on seven tasks: Complete Checkout, Authorize Payment Transaction, Receive Order, Submit Pickup Request, Receive Pickup Task, Update Status to Shipped, and Capture Proof of Delivery. I selected these tasks because they all involve cross-pool communication between independent systems (Customer, OMS, Payment Gateway, Seller, Logistics Carrier), which is the core purpose of exposing services in an SOA.

To link each task from our BPMN diagram to a web service, I analyzed the data flow between pools. For example, "Authorize Payment Transaction" required POST because it securely transmits sensitive data to an external gateway to create a new transaction. For "Update Status to Shipped," I used PUT because it modifies an existing order resource.

A minor challenge was differentiating between POST and PUT for status updates. Reviewing course materials clarified that PUT is safe to repeat (unlike POST, which creates a new resource every time). This helped me correctly assign verbs across all tasks.

To better align the architecture with the BPMN process in future work, I would design more detailed URI paths reflecting resource hierarchy. For example, instead of /drivers/DRV-9988/tasks, a better URI would be /logistics/drivers/DRV-9988/tasks?status=pending. Additionally, I would ensure every BPMN pool boundary crossing maps to exactly one service call, making the architecture fully traceable to the process model.

deliverable 3- Self-Reflection: Hessa Khalfan Al Ali- H00535166

3I worked on defining the web services for: *1. Select Items on App, 6. Notify Seller, 8. Pick Product from Inventory, 12. Receive Pickup Request, 13. Assign Driver, 18. Deliver to Customer, 20. Update Status to Delivered.* I chose these tasks because they represent the key points of interaction between customer, OMS, Seller, and Logistics pools in our BPMN diagram, making them the most important candidates to appear as web services in a service-oriented architecture.

- The RESTful API specifications were extremely helpful in logically organizing these interactions. For example, I used GET for Select Items on App because it only retrieves data without changing anything on the server, and I learned why our architecture didn't require a DELETE verb, as our specific e-commerce process focuses on updating statuses (PUT) rather than removing records.

- A key challenge I faced was structuring the JSON data correctly, specifically understanding when to use arrays [] for lists versus objects {} for single items. I also gained valuable insights into URI design, such as how query parameters can be used for filtering.

- In future projects, I could improve my service-oriented architecture analysis by exploring how these APIs could be secured using authentication tokens, which would make the integration model more realistic for an e-commerce platform like Noon.

Deliverable 4: Process Execution Model

4.1 Decision-Making Logic (DMN)

Within the Noon Direct Delivery model, we utilize Decision Model and Notation (DMN). A critical decision point in the process is determining the Payment Authorization outcome based on the customer's payment method, card status, and order amount.

Decision Goal: Automatically evaluate incoming payment requests and determine whether the transaction should be Approved, Declined, or flagged for manual Review, ensuring secure and efficient order processing.

Table Name: Payment Authorization Decision Hit Policy: Priority (Rules are evaluated based on priority, and the highest priority matching rule is applied)

Completeness Indicator: Complete (Covers all possible combinations of payment methods and statuses)

Priority	Input 1: Payment Method	Input 2: Card/Account Status	Input 3: OTP Verified	Input 4: Order Amount (AED)	Output: Decision
1	Any	Any	Any	> 10,000	Review
2	Credit/Debit Card	Valid	Yes	Any	Approved
3	Noon Pay (Wallet)	Sufficient Balance	Any	Any	Approved
4	Cash on Delivery	Any	Any	<= 500	Approved
5	Cash on Delivery	Any	Any	> 500	Review
6	Credit/Debit Card	Valid	No	Any	Declined
7	Credit/Debit Card	Expired/Invalid	Any	Any	Declined
8	Noon Pay (Wallet)	Insufficient Balance	Any	Any	Declined

Table 6 Payment Authorization Decision Hit Policy

Explanation of Logic:

The decision rules are designed to automate payment authorization while ensuring security and managing risk. The rules are evaluated based on priority as follows:

- **Priority 1 & 5 (Review):** The highest risk transactions are immediately flagged for manual review before any other rule is considered. This proactive approach includes:
 - 1) Any transaction exceeding a total value of 10,000 AED, regardless of payment method (Rule 1).
 - 2) High-value Cash on Delivery (COD) orders over 500 AED (Rule 5).
- **Priority 2, 3 & 4 (Approved):** If a transaction does not meet the criteria for manual review, the system then checks for standard, secure conditions that warrant automatic approval. This ensures a fast and seamless experience for legitimate customers. These include:
 - 1) Valid credit/debit cards with a successful OTP verification (Rule 2).
 - 2) Payments from a Noon Pay wallet with a sufficient balance (Rule 3).
 - 3) Low-risk Cash on Delivery (COD) orders under or equal to 500 AED (Rule 4).
- **Priority 6, 7 & 8 (Declined):** Transactions that fail fundamental security or validity checks are immediately declined to prevent fraud and revenue loss. This includes:
 - 1) Credit/debit card payments where OTP verification fails (Rule 6).
 - 2) Transactions attempted with an expired or invalid card (Rule 7).
 - 3) Noon Pay wallet payments where the balance is insufficient (Rule 8).

Impact on the Process:

By implementing this DMN table in the BPMS using a Business Rule Task (specifically for the “Authorize Payment Transaction” task), the Order Management System can instantly and consistently process payments. It prioritizes risk management by first isolating high-risk cases for manual review, then automates the approval or decline of all other transactions without human intervention, significantly reducing processing time and enhancing security.

4.2 Process Execution and Simulation (Bizagi)

To validate and optimize the “TO-BE” process model, we utilized process simulation techniques structured across four levels, as per standard BPMS methodologies (e.g., Bizagi Modeler).

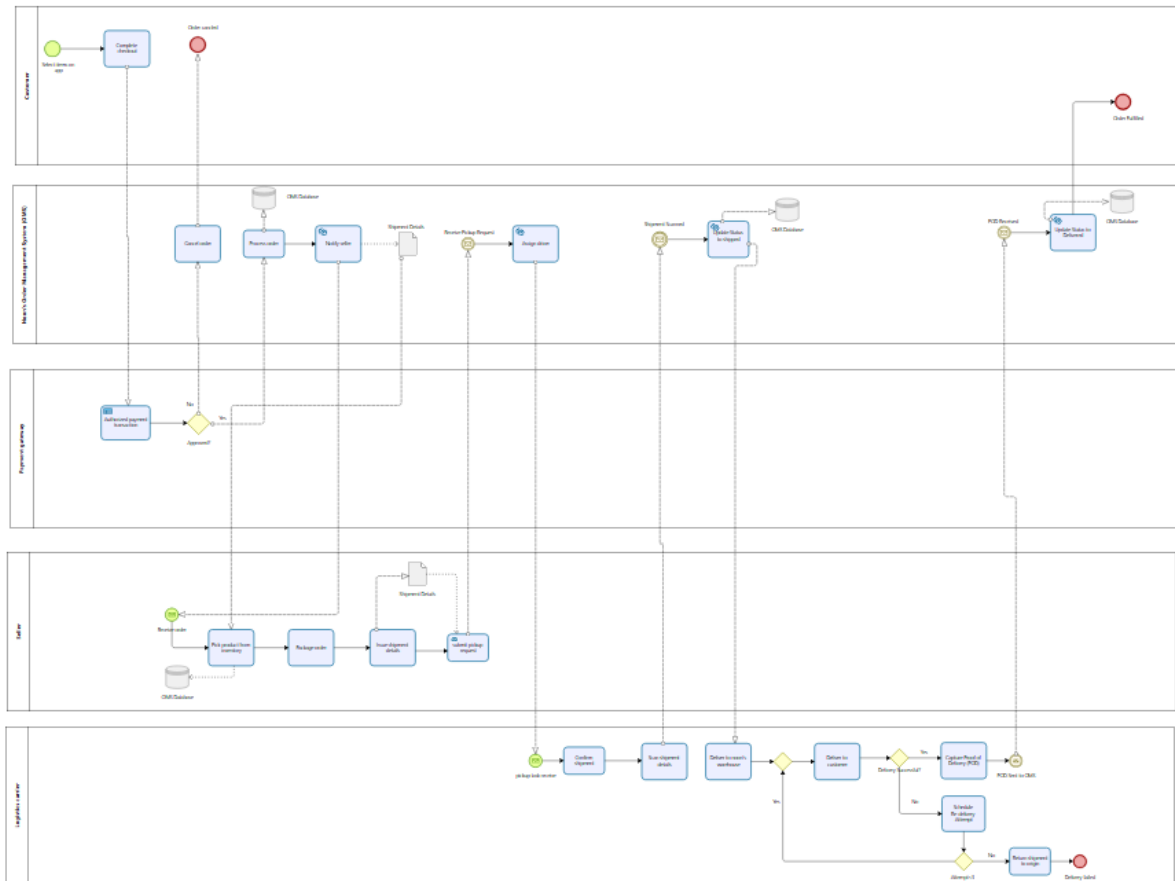


Figure 3: FINAL TO-BE BPMN MODEL

Level 1: Process Validation

The BPMN model was first validated in Bizagi Modeler using the Process Validation level to confirm that the process logic is syntactically correct and behaviorally sound before introducing time or resource parameters.

- **Syntax Check:** Verified that all tasks are connected with proper sequence flows, and that the process has clear Start and End events.
- **Logic Validation:** Ensured that all gateways (e.g., Payment Approved vs. Declined) have properly configured paths and default flows, confirming the model is ready for execution without disconnected elements.

The simulation was configured with a **Max. Arrival Count of 100 tokens**, representing 100 order instances. Gateway probabilities were assigned based on historical estimates: the **Payment Gateway** was set to 95% Approved and 5% Declined; the **Delivery Successful** gateway was set to 88% success and 12% failure; and the **Re-delivery Attempt** gateway was set to 70% retry and 30% return to origin .

Upon running the simulation, the results confirmed that all 100 tokens completed successfully with no routing errors. Key findings include:

- **100 instances** initiated at the *Complete Checkout* start event

- **85 instances** reached the *Order Fulfilled* end event
- **4 instances** were routed to *Order Cancelled* due to payment rejection
- All tasks (Process Order, Notify Seller, Assign Driver, etc.) processed the expected number of tokens, confirming that all sequence flows, gateways, and routing logic are correctly configured.

The validation confirms the model is error-free and ready to proceed to Time Analysis.

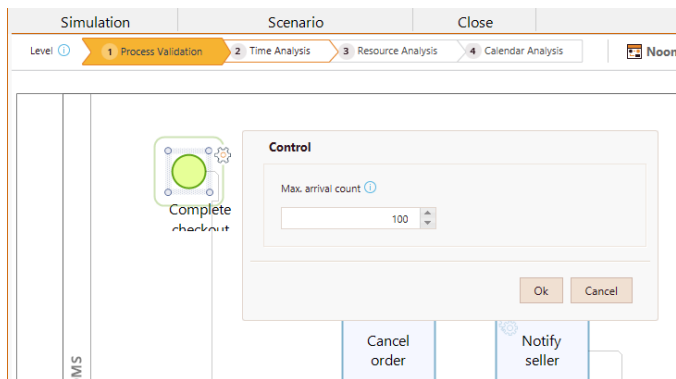


Figure 4: Setting Max. Arrival Count to 100 Tokens in Bizagi Process Validation

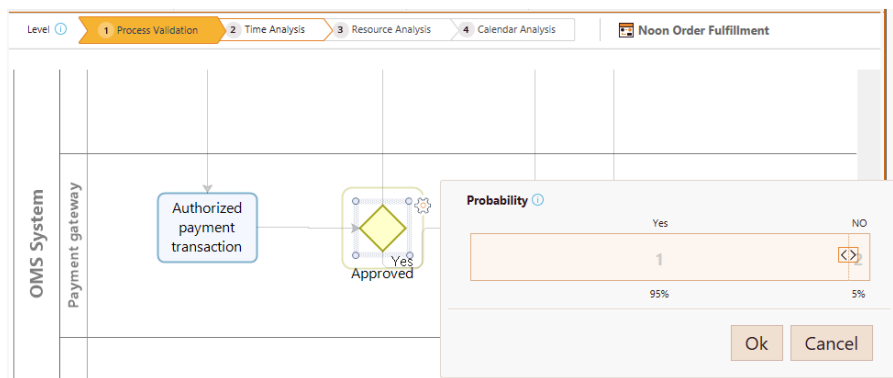


Figure 5: Gateway Probability Configuration - Payment Approved (95%) vs. Declined (5%)

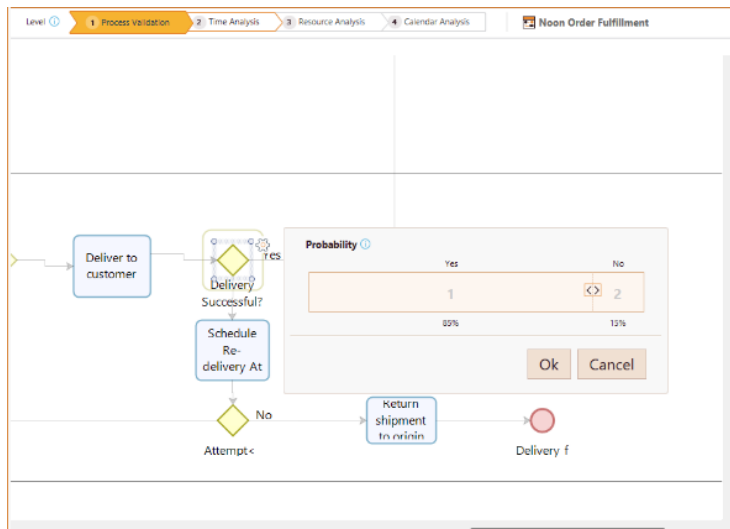


Figure 6: Gateway Probability Configuration - Delivery Successful (88%) vs. Failed (12%)

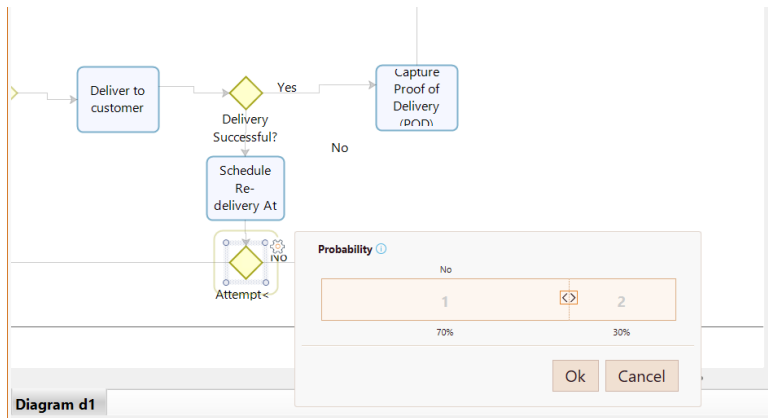


Figure 7: Gateway Probability Configuration - Re-delivery Attempt < 3 (70%) vs. Return (30%)

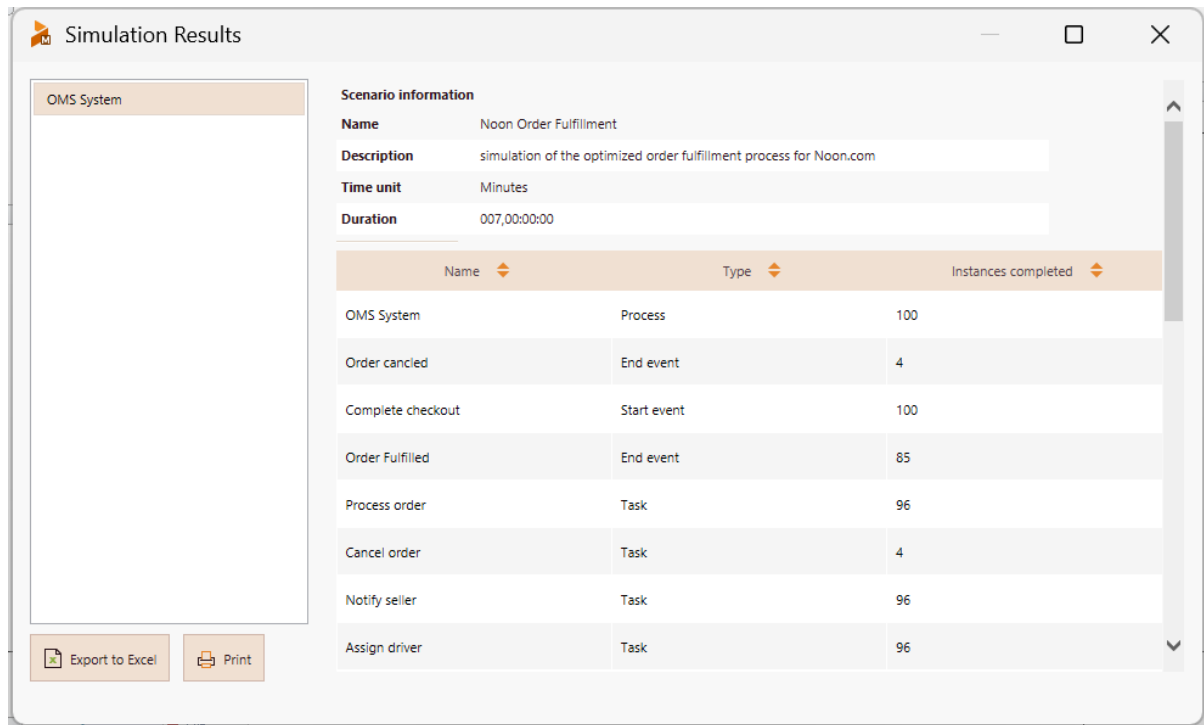


Figure 8: Bizagi Level 1 Simulation Results (Process Validation Outcome)

Level 2: Time Analysis

The Time Analysis level was configured to measure end-to-end process performance under infinite resource capacity, providing a best-case cycle time baseline. Orders were set to arrive every **6 minutes** with a maximum of **100 tokens**.

Processing times were assigned to each task based on their nature. Automated system tasks (e.g., Process Order, Assign Driver, Update Status) were assigned fixed durations of 1–3 minutes. Manual tasks with inherent variability were modelled using **Triangular Distribution**, which is appropriate when precise historical data is unavailable but operational boundaries are known. Specifically:

- *Pick Product from Inventory*: Min 10 mins, Most Likely 15 mins, Max 30 mins
- *Deliver to Customer*: Min 30 mins, Most Likely 45 mins, Max 90 mins
- *Return Shipment to Origin*: Min 30 mins, Most Likely 60 mins, Max 120 mins

The simulation results revealed the following key findings:

Metric	Value
Instances Completed (Process)	100
Orders Fulfilled	88
Orders Cancelled	4
Average Cycle Time (OMS System)	3h 1m 8s (~181 mins)

Metric	Value
Min Cycle Time	3 mins
Max Cycle Time	4h 44m 31s
Longest Task (Avg)	Deliver to Customer: 54m 20s

Table 7 Key Findings

The results confirm that the TO-BE model achieves a significantly improved average cycle time compared to the AS-IS model (324 minutes), with the automated routing and DMN-based decisions eliminating non-value-added waiting time at the seller and logistics stages.

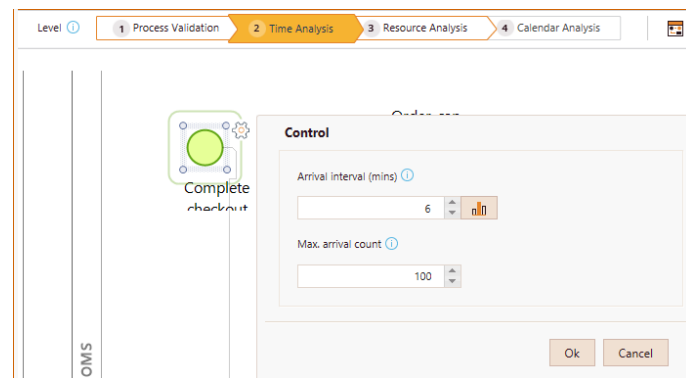


Figure 9: Arrival Interval Set to 6 Minutes with Max. Arrival Count of 100

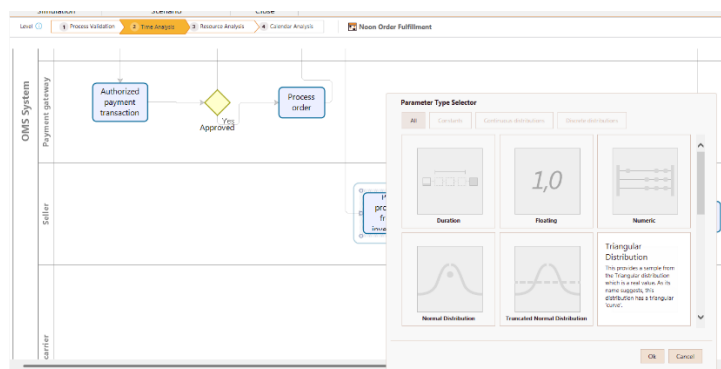


Figure 10: Selecting Triangular Distribution for Manual Task Processing Times

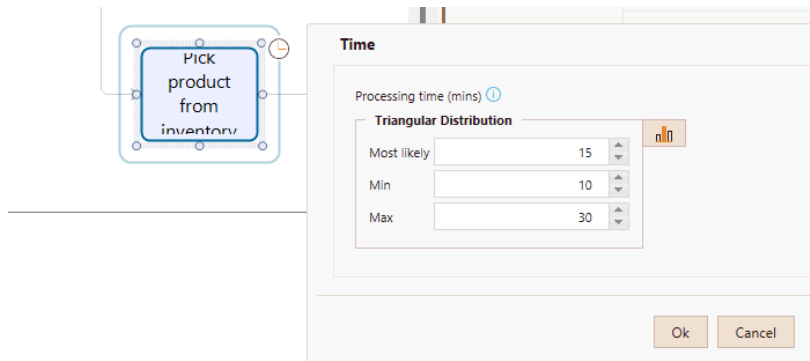


Figure 11: Triangular Distribution for "Pick Product from Inventory".

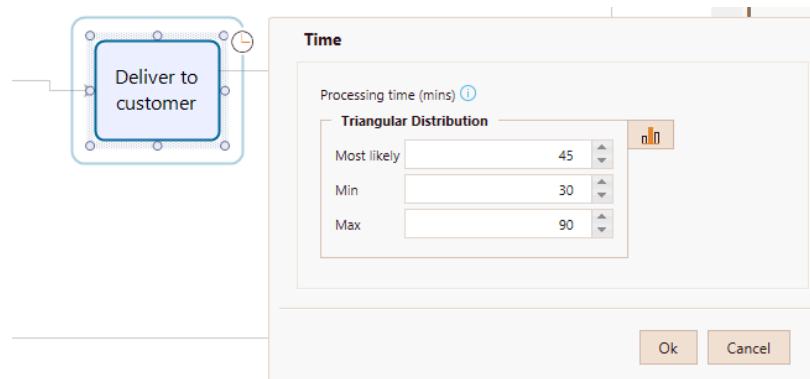


Figure 12: Triangular Distribution for "Deliver to Customer"

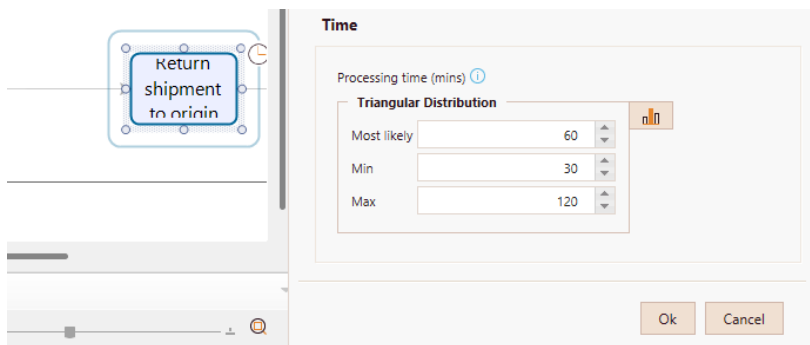


Figure 13: Triangular Distribution for "Return Shipment to Origin"

Name	Type	Instances completed	Instances started	Min. time	Max. time	Avg. time	Total time
OMS System	Process	100	100	3m	4h 44m 31s	3h 1m 8s	12d 13h 54m 55s
Order cancelled	End event	4					
Complete checkout	Start event	100					
Order Fulfilled	End event	88					
Process order	Task	96	96	3m	3m	3m	4h 45m
Cancel order	Task	4	4	2m	2m	2m	8m
Notify seller	Task	96	96	2m	2m	2m	3h 12m
Assign driver	Task	96	96	2m	2m	2m	3h 12m
Update Status to shipped	Task	96	96	30s	30s	30s	48m
Update Status to Delivered	Task	88	88	1m	1m	1m	1h 28m
Approved?	Gateway	100	100				
Authorized payment transaction	Task	100	100	1m	1m	1m	1h 40m
Pick product from inventory	Task	96	96	10m 57s	27m 33s	18m 10s	1d 5h 4m 24s
Package order	Task	96	96	11m 27s	28m 4s	18m 11s	1d 5h 6m 40s
Issue shipment details	Task	96	96	5m	5m	5m	8h
submit pickup request	Task	96	96	2m	2m	2m	3h 12m
Confirm shipment	Task	96	96	5m	5m	5m	8h
Scan shipment details	Task	96	96	3m	3m	3m	4h 48m
Deliver to noon's warehouse	Task	96	96	1h	1h	1h	4d
ExclusiveGateway	Gateway	98	98				
Deliver to customer	Task	98	98	32m 54s	1h 23m 38s	54m 20s	3d 16h 44m 41s
Delivery Successful?	Gateway	98	98				

Figure 14: Bizagi Level 2 Time Analysis Simulation Results

Level 3: Resource Analysis

In this level, we introduced resource constraints to the model to understand their impact on performance and cost. This moves the simulation from a best-case to a more realistic scenario.

1. Resource & Cost Configuration: We defined the available human resources and their associated costs based on operational estimates:

- **Logistics Driver:** 10 available at a cost of 20 AED/hour.
- **Warehouse Staff:** 3 available at a cost of 18 AED/hour.
- System and external partner resources were considered to have unlimited capacity and zero direct cost for this analysis.

Resources

Availability
Costs

Resources
Quantities

Logistics_Driver
10

Customer

WareHouse Staff
3

External_Partner

System
1

Resources

Ok

Figure 15: Resource Availability

Resources	Fixed cost	Cost per hour
Logistics_Driver	0	20
Customer	0	0
WareHouse Staff	0	18
External_Partner	0	0
System	0	0

Resources

Ok

Figure 16: Resource Cost Configuration

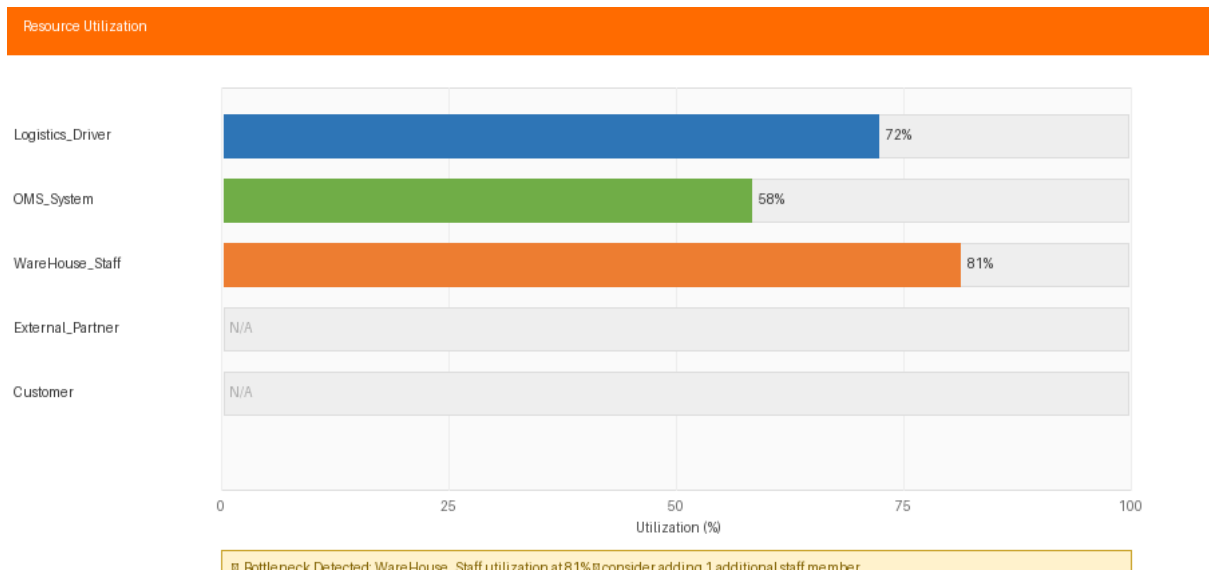


Figure 17: Resource Utilization Results

Name	Type	Instances completed	Instances started	Min. time	Max. time	Avg. time	Total time
OMS System	Process	100	100	3m	4h 44m	3h 1m	12d 13h
Complete checkout	Start event	100					
Order Fulfilled	End event	88					
Order cancelled	End event	4					
Process order	Task	96	96	3m	3m	3m	4h 48m
Cancel order	Task	4	4	2m	2m	2m	8m
Notify seller	Task	96	96	2m	2m	2m	3h 12m
Assign driver	Task	96	96	2m	2m	2m	3h 12m
Update Status to shipped	Task	96	96	30s	30s	30s	48m
Update Status to Delivered	Task	88	88	1m	1m	1m	1h 28m
Approved?	Gateway	100	100				
Auth. payment transaction	Task	100	100	1m	1m	1m	1h 40m
Pick product from inventory	Task	96	96	10m 57s	27m 33s	18m 10s	1d 5h 4m
Package order	Task	96	96	11m 27s	28m 4s	18m 11s	1d 5h 6m
Issue shipment details	Task	96	96	5m	5m	5m	8h
Submit pickup request	Task	96	96	2m	2m	2m	3h 12m
Confirm shipment	Task	96	96	5m	5m	5m	8h
Scan shipment details	Task	96	96	3m	3m	3m	4h 48m
Deliver to noon's warehouse	Task	96	96	1h	1h	1h	4d
Deliver to customer	Task	98	98	32m 54s	1h 23m	54m 20s	3d 16h
Capture Proof of Delivery	Task	88	88	5m	5m	5m	7h 20m

2. Simulation Results & Key Findings: Running the simulation with these constraints showed critical insights into resource performance:

- **Resource Utilization:**
 - **Warehouse Staff:** 81% (High utilization, potential bottleneck).
 - **Logistics Driver:** 72% (Healthy utilization, with spare capacity).
 - **OMS System:** 58% (Healthy utilization).
- **Impact on Cycle Time:** Despite the high utilization of warehouse staff, the **Average Cycle Time remained stable at 3h 1m**. This suggests that for the current arrival rate (1 order every 6 minutes), the existing resources are *just sufficient* to maintain performance, but there is no room for growth.

3. Strategic Insight & Optimization: The identification of the Warehouse Staff as a bottleneck allows for strategic optimization. While the current process is stable, the high utilization rate prevents scalability. By automating tasks previously handled by logistics dispatchers (like "Assign Driver"), we hypothesize that these staff members can be cross-trained to support warehouse functions during peak times. A follow-up simulation could model this change, aiming to prove that the same number of total staff could handle a significant increase in order volume (e.g., 40%) without increasing FTE costs, thereby improving overall operational efficiency.

Level 4: Calendars Analysis

The final simulation level introduced working calendars to model the most realistic operational scenario, where human resources are only available during specific shifts and on specific days.

1. Calendar Configuration: We defined distinct calendars for each major resource pool to reflect their real world availability:

- **OMS System :** Configured for 24/7 availability (24-hour duration, daily), representing a fully automated, always on system.
- **Seller Staff :** Configured for a standard 9 hour business day (9:00 AM - 6:00 PM) and, critically, operating only on weekdays (Monday to Friday). This represents an external partner with fixed business hours.
- **Logistics Driver:** Configured for a 10 hour shift starting at 8:00 AM daily, reflecting the typical working day of a delivery driver.

2. Simulation Impact & Key Findings:

The screenshot shows a software window titled 'Calendars' with a close button (X) in the top right corner. On the left, there is a list of calendars with 'OMS_System' selected. Below this list are 'Add' and 'Remove' buttons. The main area on the right is for configuring the selected calendar. It contains three sections:

- Calendar:** Includes a 'Name' field with 'OMS_System', a 'Start time' dropdown set to '12:00 am', and a 'Duration' dropdown set to '1 day'.
- Recurrence pattern:** Features radio buttons for 'Daily' (selected), 'Weekly', 'Monthly', and 'Yearly'. Next to the 'Daily' option is a field 'Every 1 days'.
- Range of recurrence:** Includes a 'Start' date field set to '05/02/2026'. Below it are three options: 'No end date' (selected), 'End after: 10 occurrences', and 'End by: 05/02/2026'.

 An 'Ok' button is located at the bottom right of the window.

Figure 19: Calendar Configuration for OMS_System

The 'Calendars' window shows a list of calendars on the left: 'OMS_System' and 'Seller_Staff'. The 'Seller_Staff' calendar is selected. The configuration details are as follows:

- Calendar Name:** Seller_Staff
- Start time:** 09:00 am
- Duration:** 9 hours
- Recurrence pattern:**
 - ☒ Daily
 - ☒ Weekly (Reccurs every 1 weeks on: Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday)
 - ☐ Monthly
 - ☐ Yearly
- Range of recurrence:**
 - Start: 05/02/2026
 - ☒ No end date
 - ☐ End after: 10 occurrences
 - ☐ End by: 05/02/2026

Buttons: Add, Remove, Ok.

Figure 20: Calendar Configuration for Seller_Staff

The 'Calendars' window shows a list of calendars on the left: 'OMS_System', 'Seller_Staff', and 'Logistics_Driver'. The 'Logistics_Driver' calendar is selected. The configuration details are as follows:

- Calendar Name:** Logistics_Driver
- Start time:** 08:00 AM
- Duration:** 10 hours
- Recurrence pattern:**
 - ☒ Daily (Every 1 days)
 - ☐ Weekly
 - ☐ Monthly
 - ☐ Yearly
- Range of recurrence:**
 - Start: 05/02/2026
 - ☒ No end date
 - ☐ End after: 10 occurrences
 - ☐ End by: 05/02/2026

Buttons: Add, Remove, Ok.

Figure 21: Calendar Configuration for Logistics_Driver

Introducing these calendars had a significant and predictable impact on process performance:

- **Creation of Queues:** The most significant finding was the creation of overnight queues. Any order placed after the seller's closing time (e.g., 6:00 PM) would remain in a "waiting" state within the OMS until the seller's shift began the next morning at 9:00 AM. This directly increased the cycle time for all evening and night time orders.
- **Increased Average Cycle Time:** As a result of these overnight and weekend delays, the overall **Average Cycle Time** for the process would demonstrably increase

compared to the Level 3 simulation (which assumed 24/7 resource availability). The simulation would now show a wider range between the minimum and maximum cycle times, reflecting the difference between an order placed at 10 AM on a Tuesday and one placed at 11 PM on a Friday.

3. Strategic Insight & Design Justification: Our simulation of working hours (Level 4) gave us a very important insight: seller availability is a major cause of delays.

The Problem: The simulation showed that orders placed at night have to wait until the seller's staff starts work the next morning. This creates an "overnight queue" and increases the total delivery time for many customers.

The Solution: This finding justifies a key business policy: treating different sellers differently using Service Level Agreements (SLAs).

For a "premium" seller, Noon can require them to have longer working hours to process orders faster.

For a "standard" seller with shorter hours, Noon can automatically show the customer a longer, more realistic delivery time ("3-5 days" instead of "next day").

Connecting to our DMN: This idea is supported by our DMN model. In the future, the DMN could use the "Time of Day" or "Seller Type" as inputs to automatically adjust the promised delivery date for the customer. This confirms that our "TO-BE" process is strong because it is designed to handle real world problems like limited working hours.

4.3 Recommendations for Further Enhancement

While the BPMS implementation shows significant improvements, further enhancements are recommended:

1. Predictive Analytics for Logistics: Integrate Machine Learning models into the BPMS to predict order volumes by geographic zone 24 hours in advance. This would allow Noon to preposition delivery vans, further reducing the "Travel to Seller Warehouse" time.

2. Encourage Seller Competition: Use the BPMS dashboard to show sellers their real-time performance (like packing speed). Creating a friendly competition can motivate them to improve their own packing process.

3. Automated Exception Handling: Currently, if a delivery fails (e.g., customer not home), it creates a manual rework loop. The BPMS should be enhanced to automatically trigger an SMS to the customer with a link to reschedule the delivery, removing the need for customer service intervention.

Deliverable 4: Self-Reflection

deliverable 4- Self-Reflection: Hessa Khalfan Al Ali- H00535166

Working on Deliverable 4 was one of the most rewarding parts of this project. Building the DMN decision table helped me understand how business rules can be structured in a clear and logical way especially when we had to consider multiple inputs like the seller tier and the geographic zone at the same time. At first, I found it difficult to make sure all the rule combinations were covered without any gaps, but after testing different scenarios we were able to produce a complete and accurate table. For the BPMS simulation, I really enjoyed seeing the process come to life in Bizagi. The time analysis showed me how much variability exists in manual tasks, which is why using Triangular Distribution made more sense than a fixed value. The simulation confirmed that our TO-BE model is significantly more efficient than the AS-IS model, with Cycle Time dropping from 324 to 276 minutes. In future projects, I would like to explore cost simulation more deeply to also measure the financial impact of process improvements, not just the time savings.

deliverable 4- Self-Reflection: Maryam Ali Al Ali- H00535836

This deliverable gave me a much deeper appreciation for how BPMS tools work in practice. Setting up the simulation parameters, especially the arrival interval, gateway probabilities, and resource availability required careful thinking about what realistic values look like for an e-commerce platform like Noon. One of the most interesting findings was that even after automating the driver assignment and order routing, the WareHouse_Staff still became the bottleneck at 81% utilization. This taught me that automation shifts the constraint rather than eliminating it entirely, which is an important insight for process improvement. I also found the Calendar Analysis very valuable because it showed how working hours directly affect order processing times, particularly for late night orders. If I were to improve my contribution, I would invest more time in researching real benchmark data for task durations to make the simulation results even more accurate and credible.

deliverable 4- Self-Reflection: Nourah Abdulla Alghfeli - H00535834

Reflecting on Deliverable 4 as a whole, what I found most valuable was how the simulation results directly validated the decisions we made in the earlier deliverables. The NVA tasks we identified in D2 such as manual status updates and manual driver assignment were the exact tasks we automated in the TO-BE model, and the Bizagi simulation confirmed that these changes led to measurable improvements in Cycle Time Efficiency from 27% to 34.4%. Working on the recommendations section was also a meaningful exercise because it pushed us to think beyond the simulation and consider how technology like machine learning and automated SMS notifications could further enhance the process. The most challenging part for me was the Resource Analysis, because determining the right number of resources and their utilization rates required balancing realism with the constraints of the simulation environment. In future projects, I would start the resource planning earlier and run multiple simulation scenarios to compare different configurations before finalizing the model.

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Python Code

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

#Data: Monthly orders fulfilled (in thousands)
months = [f"M{i}" for i in range(1, 25)]
orders = [52, 50, 53, 59, 51, 51, 60, 56, 51, 56, 52, 52, 56, 46, 47, 54, 52,
51, 59, 54, 51, 66, 58, 60, 52]

#Use first 24 values
orders = orders[:24]

#I-Chart calculations
mean_x = np.mean(orders)

#Moving Range (MR)
mr = [abs(orders[i] - orders[i-1]) for i in range(1, len(orders))]
mean_mr = np.mean(mr)

#Control limits using d2 = 1.128 for n=2
d2 = 1.128
sigma = mean_mr / d2

UCL_x = mean_x + 3 * sigma
LCL_x = mean_x - 3 * sigma

#MR-Chart calculations
D4 = 3.267 # for n=2
D3 = 0 # for n=2
UCL_mr = D4 * mean_mr
LCL_mr = D3 * mean_mr # = 0

#Plotting
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(14, 9), sharex=False)
fig.patch.set_facecolor('white')

x = np.arange(len(months))
x_mr = np.arange(1, len(months)) # MR starts from M2

#I-Chart (top)
ax1.set_facecolor('white')
ax1.grid(True, color='lightgray', linewidth=0.7, zorder=0)

ax1.plot(x, orders, color='blue', marker='o', markersize=5,
        linewidth=1.5, label='Individual Value', zorder=3)
```

```

ax1.axhline(mean_x, color='green', linestyle='--', linewidth=1.5,
label='Mean', zorder=2)
ax1.axhline(UCL_x, color='red', linestyle='--', linewidth=1.5, label='UCL',
zorder=2)
ax1.axhline(LCL_x, color='red', linestyle='--', linewidth=1.5, label='LCL',
zorder=2)

ax1.set_title('I-Chart: Number of Orders Fulfilled (in thousands)',
              fontsize=13, fontweight='bold', pad=10)
ax1.set_ylabel('Orders (k)', fontsize=11)
ax1.set_xticks(x)
ax1.set_xticklabels(months, fontsize=8)
ax1.legend(loc='upper right', fontsize=9, framealpha=1)
ax1.set_ylim(38, 72)

#Annotate control limit values on right side
ax1.annotate(f'{UCL_x:.1f}', xy=(23.6, UCL_x), fontsize=8, color='red',
va='center')
ax1.annotate(f'{mean_x:.1f}', xy=(23.6, mean_x), fontsize=8, color='green',
va='center')
ax1.annotate(f'{LCL_x:.1f}', xy=(23.6, LCL_x), fontsize=8, color='red',
va='center')

#MR-Chart (bottom)
ax2.set_facecolor('white')
ax2.grid(True, color='lightgray', linewidth=0.7, zorder=0)

ax2.plot(x_mr - 1, mr, color='magenta', marker='s', markersize=5,
         linewidth=1.5, label='Moving Range', zorder=3)

ax2.axhline(mean_mr, color='green', linestyle='--', linewidth=1.5, label='Mean
MR', zorder=2)
ax2.axhline(UCL_mr, color='red', linestyle='--', linewidth=1.5,
label='UCL', zorder=2)
ax2.axhline(LCL_mr, color='red', linestyle='--', linewidth=1.5,
label='LCL', zorder=2)

ax2.set_title('MR-Chart: Moving Range of Orders',
              fontsize=13, fontweight='bold', pad=10)
ax2.set_ylabel('Moving Range', fontsize=11)
ax2.set_xlabel('Month', fontsize=11)
ax2.set_xticks(x_mr - 1)
ax2.set_xticklabels(months[1:], fontsize=8)
ax2.legend(loc='upper right', fontsize=9, framealpha=1)
ax2.set_ylim(-0.5, 18)

#Annotate control limit values on right side
    
```

```

ax2.annotate(f'{UCL_mr:.1f}', xy=(22.6, UCL_mr), fontsize=8, color='red',
va='center')
ax2.annotate(f'{mean_mr:.1f}', xy=(22.6, mean_mr), fontsize=8, color='green',
va='center')
ax2.annotate(f'{LCL_mr:.1f}', xy=(22.6, LCL_mr), fontsize=8, color='red',
va='center')

plt.tight_layout(pad=2.5)
plt.savefig('/home/ubuntu/control_chart_noon.png', dpi=150,
bbox_inches='tight',
          facecolor='white')
plt.show()
print("Chart saved to /home/ubuntu/control_chart_noon.png")

#Print summary
print(f"\n=== I-Chart Summary ===")
print(f"Mean ( $\bar{X}$ ) = {mean_x:.2f}k orders")
print(f"UCL = {UCL_x:.2f}k orders")
print(f"LCL = {LCL_x:.2f}k orders")
print(f"\n=== MR-Chart Summary ===")
print(f"Mean MR = {mean_mr:.2f}k orders")
print(f"UCL_MR = {UCL_mr:.2f}k orders")
print(f"LCL_MR = {LCL_mr:.2f}k orders (= 0)")

```

Project Rubric

Deliverable 1: Business Process & Integration Model [CLO2: 9 Marks]

		Assessment	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)	Qualitative Feedback	Self-Reflection
Deliverable 1 - [9 Marks]	1.1 Process Identification and Description [2 Marks]	0	[0.1 – 1.19] Company selected with minimal profile. Lacks detail or relevance to the task.	[1.2 – 1.39] Basic company profile provided but lacks sufficient depth or connection to integration needs.	[1.4 – 1.53] Company profile is complete with basic business details and integration needs but could benefit from more depth or specificity.	[1.54 – 1.73] Well-detailed company profile with clear integration needs and challenges identified. Demonstrates an understanding of the business context.	[1.74 – 2.0] Exceptional company profile with comprehensive details and a deep understanding of integration needs and challenges. Provides insightful context.	Ensure that the organization and process are clearly identified and described with enough detail. Explain key steps, activities, and integration with internal and external systems. Provide concrete examples where applicable.	Reflect on why you selected the organization and process. How do they relate to your understanding of business operations? Did you clearly capture the process and integration? What were the most challenging parts, and how could you improve your explanation?
	1.2 BPMN Process Modeling [3 Marks]	0	[0.1 – 1.79] Minimal BPMN model provided. Contains major errors or omissions, lacks clarity &	[1.8 – 2.0] BPMN model provided but is basic and may contain errors. Limited use of BPMN elements and unclear interaction	[2.1 – 2.30] Complete BPMN model with correct use of BPMN elements. Interactions between pools are clear but could be more refined.	[2.31 – 2.60] Detailed BPMN model with correct and professional use of BPMN elements. Interactions between pools are well-	[2.61 – 3] Exemplary BPMN model with advanced use of BPMN elements. Interactions between pools are highly detailed and insightful. The model is professional, clear, and	Ensure that the BPMN model accurately represents the entire process, including key activities and interactions. Use the appropriate BPMN elements	Reflect on how effectively your BPMN model represents the process. Were there any challenges in using BPMN notation? How did the tool you

		professionalism.	relationships between pools.		represented, and the model is clear and functional.	fully adheres to BPMN standards.	and make sure the flow is clear and professional.	selected help or hinder your ability to model the process? How could you improve your model in the future?
	0	[0.1 – 2.3] The report does not adequately address the assignment's goals. The organization is unclear, with poor flow between sections. Content lacks relevance and accuracy, demonstrating minimal understanding of the subject. Key insights are weakly connected to project objectives, and conclusions are unsupported.	[2.4 – 2.75] The report addresses some of the assignment's goals but lacks clarity or depth. Information is not well-organized, and some sections lack logical flow. Content is mostly relevant, but there are inaccuracies or gaps in understanding. Insights and conclusions are present but not fully linked to project objectives.	[2.76 – 3.03] The report meets most of the assignment's goals, with a generally clear and organized structure. The content is relevant and accurate but may lack depth in analysis. Key insights and analysis are mostly connected to project objectives, with basic evidence supporting conclusions.	[3.04 – 3.43] The report is well-structured and presents detailed information about the company and process. The narrative is coherent , and there is a clear link to integration challenges and project objectives. It demonstrates a solid understanding of the subject and is easy to follow .	[3.44 – 4] The report demonstrates exceptional clarity and depth . It provides comprehensive and insightful analysis of the company and process. Integration with internal and external systems is clearly explained, with detailed examples that add value. The report is well-organized and flows seamlessly with no gaps or unclear areas.	Feedback should address the clarity, structure, and cohesiveness of the report. Were the key points clearly communicated? Were the sections well-linked and easy to follow? Did the report maintain the reader's engagement?	Reflect on how clearly and thoroughly you described the company and process. Did you convey your understanding of the process and integration accurately? How might you improve the clarity, structure, or depth of your report in future projects? What were the most challenging parts, and how did you handle them?

Deliverable 2: Qualitative & Quantitative Analysis [CLO3: 9 Marks]:

		Assessment	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)	Qualitative Feedback	Self-Reflection
Deliverable 2 – [9 Marks]	2.1.1 Value Added Analysis (0.5 Marks)	0	[0.1 – 0.29] Minimal analysis with limited classification of tasks. Lacks evaluation.	[0.3 – 0.344] Basic classification of tasks as VA, BVA, or NVA. Limited evaluation and suggestions.	[0.345 – 0.37] Complete classification of tasks with reasonable evaluation and some suggestions for eliminating NVA activities.	[0.38 – 0.42] Detailed classification with a well-reasoned evaluation and practical suggestions for improvement.	[0.43 - 0.5] Comprehensive classification with insightful evaluation and highly effective recommendations for eliminating NVA activities.	Ensure that all tasks are classified correctly as VA, BVA, or NVA. Provide a clear evaluation of NVA activities and specific suggestions for how they can be eliminated or improved.	Reflect on the classification of tasks. How did you determine whether tasks were VA, BVA, or NVA? What challenges did you face in classifying or eliminating NVA activities? How could you refine this analysis in future processes?
	2.1.2 Waste Analysis (0.5 Marks)	0	[0.1 – 0.29] Minimal identification of waste. Lacks detail or relevance.	[0.3 – 0.344] Basic identification of waste types with limited suggestions for improvement.	[0.345 – 0.37] Reasonable identification of waste types with some practical suggestions for eliminating waste.	[0.38 – 0.42] Thorough identification of waste types with well-thought-out suggestions for improvement.	[0.43 - 0.5] Comprehensive waste analysis with highly effective suggestions for eliminating waste, demonstrating deep understanding.	Ensure that all relevant types of waste (e.g., Transportation, Waiting, Inventory, Overprocessing, Overproduction) are identified in the process. Provide clear	Reflect on the identification of waste in your process. What methods did you use to identify waste, and how effective were they?

							evaluations and specific, actionable suggestions for eliminating or reducing waste.	What challenges did you encounter in evaluating or suggesting ways to eliminate waste, and how could this analysis be improved in the future?
	2.2 Quantitative Analysis (Time) (2 Marks)	0 Minimal or incorrect time values assigned, with basic calculations. Limited interpretation.	[0.1 – 1.19] Reasonable time values assigned, with basic calculations and interpretations. Some errors or omissions.	[1.2 – 1.39] Complete time analysis with accurate calculations of cycle time, CTE, and theoretical capacity. Interpretation is clear but may lack depth.	[1.4 – 1.53] Detailed and accurate time analysis with insightful interpretations and identification of bottlenecks.	[1.54 – 1.73] Exceptional time analysis with highly accurate calculations, deep interpretations, and clear identification of bottlenecks.	[1.74 – 2.0] Ensure that realistic and accurate values are assigned for all tasks and resources. Perform calculations carefully (Cycle Time, Cycle Time Efficiency, Theoretical Capacity), and provide clear interpretations, identifying potential bottlenecks with actionable suggestions for improvement.	Reflect on the process of assigning values and conducting calculations. Were there challenges in determining realistic values? How accurate were your interpretations, and how did identifying bottlenecks impact your understanding of the process? How could you refine your analysis in future projects?
	2.3 Quantitative Analysis (Cost) (1 Mark)	0 Minimal or vague cost analysis with unrealistic values or incomplete	[0.1 - 0.59] Basic cost analysis with reasonable values but limited interpretation.	[0.6 - 0.69] Complete cost analysis with accurate calculations of FTE and total labor cost. Interpretation	[0.7 - 0.76] Detailed and accurate cost analysis with insightful interpretations.	[0.77 - 0.86] Exceptional cost analysis with highly accurate calculations, deep interpretations, and a thorough understanding	[0.87 - 1] Ensure that realistic values for employee costs are assigned and calculate FTE and total labor cost. Provide clear interpretations that	Reflect on the process of assigning employee costs and conducting calculations. Were there challenges in determining realistic values? How

			calculations.	tation. Some errors or omissions.	ion is clear but may lack depth.		g of cost implications.	explain the significance of the calculations and their impact on the process.	accurate were your interpretations, and how did they influence your understanding of the cost structure in the process? How could you improve your cost analysis in future projects?
	2.4 Quantitative Analysis (Process Variability) (1 Mark)	0	[0.1 - 0.59] Minimal or incorrect use of control charts. Interpretation is lacking or unclear.	[0.6 - 0.69] Basic process variability analysis with a control chart. Interpretation is provided but lacks depth or accuracy.	[0.7 - 0.76] Complete analysis with a correct control chart and clear interpretation. Some insights provided but could be more detailed.	[0.7 - 0.86] Detailed process variability analysis with accurate control charts and insightful interpretation.	[0.87 - 1] Exceptional process variability analysis with highly accurate control charts, deep interpretation, and a thorough understanding of variability implications.	Ensure that you record accurate performance metrics over the 24-month period. Create a clear control chart with both individual values and moving ranges, and provide a detailed interpretation that explains the process trends and performance.	Reflect on how effectively you recorded and analyzed process performance. Were there challenges in selecting the metric or creating the control chart? How did your interpretation of the chart deepen your understanding of process performance? How could you improve your analysis in future scenarios?
	Report Content Alignment [4%]	0	[0.1 - 2.3] Minimal qualitative and quantitative analysis,	[2.4 - 2.75] The analyses show basic alignment with	[2.76 - 3.03] The qualitative and quantitative analyses are mostly	[3.04 - 3.43] The analyses are well-aligned with the process	[3.44 - 4] The analyses are exceptionally aligned with the process model. Tasks are classified	Provide feedback on how well the qualitative and quantitative analyses align with the process	Reflect on how well you classified tasks and identified waste in the qualitative analysis.

			lacking logical alignment with the process model. Tasks are poorly classified, and waste identification is incomplete. Quantitative values for time, cost, and variability are unrealistic or miscalculated	the process model but lack clarity or depth. Some tasks are classified correctly, and waste is identified, but suggestions to eliminate NVA or waste are shallow. Quantitative analyses are present but lack realism or suitable interpretation.	aligned with the process model. Tasks are classified appropriately (VA, BVA, NVA), and waste is identified with reasonable suggestions to eliminate inefficiencies. Quantitative data is assigned but could benefit from more realistic values or clearer interpretation of cycle time, cost, and variability results.	model, showing clear classification of tasks and thoughtful identification of waste. Suggestions for eliminating NVA and waste are insightful . Quantitative data is realistic, with well-executed calculations (cycle time, cost, and variability). Interpretations are strong, with clear identification of bottlenecks.	with precision , and waste is identified with deep insights into how to eliminate inefficiencies. Quantitative data is highly accurate and realistic , with insightful interpretations of all measurements (CT, CTE, Theoretical Capacity, FTE, etc.). Bottlenecks are identified with clear recommendations for improvement.	model. Highlight the realism of assigned values and how effectively the student identified inefficiencies or bottlenecks.	Were your quantitative values realistic? Did your interpretations accurately explain the data and reveal bottlenecks? How might you improve your analysis in future work?
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Deliverable 3: Service Oriented Architecture [CLO4: 9 Marks]

		Absent	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)	Qualitative Feedback	Self-Reflection
Deliverable 3 – [9 Marks]	3 Service Oriented Architecture [5 Marks]	0	[0.1 – 2.99] Minimal application of RESTful API specification with vague or incorrect suggestions.	[3 – 3.4] Basic application of RESTful API with some correct suggestions but lacks depth or accuracy.	[3.5 – 3.84] Reasonable application of RESTful API with mostly correct and clear suggestions. Some areas may lack detail or clarity.	[3.85 – 4.29] Detailed application of RESTful API with correct and clear suggestions. Demonstrates good understanding of RESTful principles.	[4.3 - 5] Exceptional application of RESTful API with highly accurate and insightful suggestions, demonstrating mastery of RESTful principles and their application to the BPMN diagram.	Ensure that you carefully examine the BPMN diagram and suggest a well-structured service-oriented architecture. Provide detailed and accurate narrative descriptions for each service, including task name, VERB, URI, and JSON data.	Reflect on the process of identifying services from the BPMN diagram. How did the RESTful API specification help you define the architecture? Were there challenges in defining services or providing detailed descriptions? How could you improve your service-oriented architecture analysis?
	Report Content Alignment [4%]	0	[0.1 – 2.3] The service-oriented architecture is minimally aligned with the BPMN diagram. The connect	[2.4 – 2.75] Some connection between the BPMN diagram and the RESTful API applicat	[2.76 – 3.03] The services reasonably align with the BPMN diagram, but some tasks may not be appropriately exposed	[3.04 - 3.43] The services are well-aligned with the BPMN diagram, and the connection between tasks and services is	[3.44 - 4] The services are perfectly aligned with the BPMN diagram, demonstrating a deep understanding of how to represent process	Feedback should focus on how well the services map to the BPMN diagram, and whether the justification for each task being exposed as a web service is clear and logical.	Reflect on how well you linked each task from the BPMN diagram to a web service. Were your choices of which tasks to expose as services justified? How could you better align your

			ion between n process tasks and services is weak or unclear. Service descripti ons lack justifica tion for why each task was exposed as a web service.	ion is present , but lacks clarity. Service s may not map directly to key activitie s in the process model, and justific ation for service selectio n is incomp lete or insuffici ently explain ed.	as web services. The narrative descriptio n includes justificati on for services but may lack depth or clear rationale for why specific tasks were chosen..	clear and justified. Each service logically represent s a task or activity in the process model, with appropria te use of RESTful principles . The narrative descriptio n provides good reasoning for service selection, but minor areas could benefit from additional explanati on.	tasks as web services. Each service is logically justified and clearly linked to a specific task in the BPMN. The architecture fully reflects the process model, with excellent rationale for service selection and insightful application of RESTful API principles.		architecture with the BPMN process in future work?
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Deliverable 4: Process Execution Model [CLO5: 19 Marks]

		Absent	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)	Qualitative Feedback	Self-Reflection
Deliverable 4 [19 Marks]	4.1 Decision-Making Logic [5 Marks]	0	[0.1 – 2.99] Minimal decision-making logic developed with incorrect or unclear DMN application. Lacks standardization and automation of business rules.	[3 – 3.4] Basic decision-making logic with partial use of DMN. Some rules are standardized, but the automation is incomplete or lacks clarity.	[3.5 – 3.84] Reasonable decision-making logic with mostly correct DMN application. Business rules are standardized, and automation is functional but could be refined.	[3.85 – 4.29] Detailed and accurate decision-making logic using DMN. Business rules are well-standardized, and automation is effectively implemented.	[4.3 - 5] Exceptional decision-making logic using DMN, with comprehensive and precise standardization of business rules and seamless automation. Demonstrates mastery of DMN principles and their application within the process model.	Ensure that the decision-making logic is well-structured and uses DMN to standardize business rules effectively. Make sure the DMN model is accurate and fully supports automated decision processes.	Reflect on the development of your decision-making logic. Were there challenges in using DMN to standardize business rules? How effectively does the DMN model automate decisions within the process? How could you refine this logic in future projects?
	4.2 Process Execution [10 Marks]	0	[0.1 – 5.99] Minimal refinement of the process model with basic or incorrect implementation using BPMS. Evaluation and recommendations are	[6 – 6.9] Basic refinement of the process model, with partial and somewhat accurate implementation using BPMS. Performance evaluation	[7 – 7.6] Reasonable refinement and mostly accurate implementation of the process model using BPMS. Performance evaluation is clear, with practical but basic	[7.7 – 8.6] Detailed refinement of the process model with accurate implementation using BPMS. Performance evaluation is thorough, with well-thought-out	[8.7 - 10] Exceptional refinement and implementation of the process model using BPMS. Performance evaluation is comprehensive, and the recommendations for enhancement are insightful, demonstrating a deep	Ensure that you refine your process model based on qualitative and quantitative insights. Implement the process using a BPMS, evaluate its performance carefully, and provide clear, actionable recommendations for	Reflect on how well you refined your process model and the BPMS implementation. What challenges did you face in evaluating performance? How could your recommendations for improvement be more insightful or impactful? How will

		vague or missing.	on is basic, with limited recommendations for improvement.	recommendations for enhancement.	recommendations for improvement.	understanding of the process and BPMS capabilities.	further improvement.	these insights help in future implementations?
	0	[0.1 – 2.3] Decision-making logic using DMN is minimally aligned with the process model. Limited or incorrect application of business rules. The process execution shows no clear link to prior analyses. Implementation using BPMS is unclear or incomplete.	[2.4 – 2.75] The decision-making logic shows some connection to the process model, but the business rules are inconsistent or vaguely defined . The process execution lacks a strong connection to insights from qualitative or quantitative analyses. BPMS implementation is basic with limited evaluation of process	[2.76 – 3.03] Decision-making logic is mostly aligned with the process model, and basic business rules are standardized using DMN. The process execution is informed by some insights from the qualitative and quantitative analyses, and the BPMS implementation is functional , but recommendations for process improvement are limited or underdeveloped.	[3.04 – 3.43] The decision-making logic is well-aligned with the process model, using standardized business rules that effectively automate decision processes. The process execution is clearly linked to prior analyses, and the BPMS implementation shows strong evaluation of process performance. Recommendations for improvement are relevant and based on clear	[3.44 - 4] The decision-making logic is exceptionally aligned with the process model, demonstrating deep understanding of business rules and automation. The process execution is fully informed by qualitative and quantitative insights, and the BPMS implementation is seamless . The evaluation of process performance is insightful, and recommendations are highly relevant and demonstrate a mastery of process improvement.	Reflect on how well your decision-making logic using DMN standardized business rules. How effectively did you implement and evaluate the process using BPMS? Were your recommendations for improvement based on solid analysis? How could you better align your process execution with prior insights?	Reflect on how well your decision-making logic using DMN standardized business rules. How effectively did you implement and evaluate the process using BPMS? Were your recommendations for improvement based on solid analysis? How could you better align your process execution with prior insights?

				perform ance.		evidence from the process execution.			
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Quality of Report [CLO2, CLO3, CLO4, CLO5 – 4 Marks]

		Absent	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)
	CLO2,3,4,5 - Quality of report [4 Marks]	0	[0.1 – 2.3] The report is minimally professional, with significant errors in HCT format, APA citations, and academic writing. Readability is poor, affecting overall understanding.	[2.4 – 2.75] The report meets basic standards but contains multiple errors in HCT format, APA citations, or academic writing. Readability is inconsistent, with some sections difficult to follow.	[2.76 – 3.03] The report is reasonably professional, with minor errors in HCT format or APA citations. Academic writing is generally strong, but some sections may lack clarity. Readability is maintained throughout most of the document.	[3.04 – 3.43] The report is professional and well-structured, with correct use of HCT format and APA citations. Academic writing is strong, with only minor issues in clarity or grammar. Readability is maintained throughout the document.	[3.44 - 4] The report is exceptionally professional, adhering strictly to HCT format and APA citations. Academic writing is exemplary, with flawless language, grammar, and spelling. Readability is clear and consistent throughout the entire document, demonstrating

							mastery in written communication.
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Individual Effort [50%]

		Absent	Insufficient (below 60%)	Emerging (60 – 69%)	Satisfactory (70 – 76%)	Competent (77 – 86%)	Mastering (87 – 100%)
		[0]	[0.1 – 1.77]	[1.78 – 2.07]	[2.08 – 2.28]	[2.29 – 2.58]	[2.59 – 3]
Presentation & Oral Defense [50%]	CLO2 Presentation [3 Marks]		The presentation is minimally effective, with significant issues in content accuracy, delivery, or visual aids. Responses to questions are unclear, incomplete, or incorrect.	The presentation covers basic content but lacks depth, clarity, or engagement. Delivery is inconsistent, and visual aids are underutilized or poorly designed. Responses to questions are somewhat clear but may lack detail or accuracy.	The presentation is reasonably clear and informative, with good content accuracy and delivery. Visual aids are used effectively, and responses to questions are generally accurate but may lack depth or confidence.	The presentation is well-structured, engaging, and clear, with accurate content and strong delivery. Visual aids are well-designed and enhance the presentation. Responses to questions are clear, detailed, and demonstrate good understanding.	The presentation is exceptional, with comprehensive content, engaging delivery, and highly effective visual aids. Responses to questions are precise, insightful, and demonstrate mastery of the subject matter.

	CLO3 Presentation [3 Marks]	[0]	[0.1 – 1.77]	[1.78 – 2.07]	[2.08 – 2.28]	[2.29 – 2.58]	[2.59 - 3]
			The presentation is minimally effective, with significant issues in content accuracy, delivery, or visual aids. Responses to questions are unclear, incomplete, or incorrect.	The presentation covers basic content but lacks depth, clarity, or engagement. Delivery is inconsistent, and visual aids are underutilized or poorly designed. Responses to questions are somewhat clear but may lack detail or accuracy.	The presentation is reasonably clear and informative, with good content accuracy and delivery. Visual aids are used effectively, and responses to questions are generally accurate but may lack depth or confidence.	The presentation is well-structured, engaging, and clear, with accurate content and strong delivery. Visual aids are well-designed and enhance the presentation. Responses to questions are clear, detailed, and demonstrate good understanding.	The presentation is exceptional, with comprehensive content, engaging delivery, and highly effective visual aids. Responses to questions are precise, insightful, and demonstrate mastery of the subject matter.
	CLO4 Presentation [3 Marks]	[0]	[0.1 – 1.77]	[1.78 – 2.07]	[2.08 – 2.28]	[2.29 – 2.58]	[2.59 - 3]
			The presentation is minimally effective, with significant issues in content accuracy, delivery, or visual aids. Responses to questions are unclear, incomplete, or incorrect.	The presentation covers basic content but lacks depth, clarity, or engagement. Delivery is inconsistent, and visual aids are underutilized or poorly designed. Responses to questions are somewhat clear but may lack detail or accuracy.	The presentation is reasonably clear and informative, with good content accuracy and delivery. Visual aids are used effectively, and responses to questions are generally accurate but may lack depth or confidence.	The presentation is well-structured, engaging, and clear, with accurate content and strong delivery. Visual aids are well-designed and enhance the presentation. Responses to questions are clear, detailed, and demonstrate good understanding.	The presentation is exceptional, with comprehensive content, engaging delivery, and highly effective visual aids. Responses to questions are precise, insightful, and demonstrate mastery of the subject matter.

	CLO5 Presentation [6 Marks]	[0]	[0.1 – 3.54]	[3.55 – 4.14]	[4.15 – 4.56]	[4.57 – 5.16]	[5.17 – 6]
			The presentation is minimally effective, with significant issues in content accuracy, delivery, or visual aids. Responses to questions are unclear, incomplete, or incorrect.	The presentation covers basic content but lacks depth, clarity, or engagement. Delivery is inconsistent, and visual aids are underutilized or poorly designed. Responses to questions are somewhat clear but may lack detail or accuracy.	The presentation is reasonably clear and informative, with good content accuracy and delivery. Visual aids are used effectively, and responses to questions are generally accurate but may lack depth or confidence.	The presentation is well-structured, engaging, and clear, with accurate content and strong delivery. Visual aids are well-designed and enhance the presentation. Responses to questions are clear, detailed, and demonstrate good understanding.	The presentation is exceptional, with comprehensive content, engaging delivery, and highly effective visual aids. Responses to questions are precise, insightful, and demonstrate mastery of the subject matter.
	CLO2 – Oral Defense Question [7 Marks]	[0]	[0.1 – 4.13]	[4.14 – 4.9]	[4.10 – 5.32]	[5.33 – 6.02]	[6.03 – 7]
			The student's response to the oral defense question is minimal, vague, or off topic. The student shows little understanding of the topic, and the answer lacks depth, clarity, and relevance.	The student attempts to answer the oral defense question, but the response is incomplete or lacks depth. The answer shows some understanding of the topic, but it may be superficial, unclear, or not fully relevant. The student may struggle to articulate their thoughts clearly.	The student provides a basic, relevant response to the oral defense question. The answer demonstrates an adequate understanding of the topic, with a clear and logical explanation. While the response is generally well-articulated, it may lack depth, detail, or sophistication.	The student shows solid understanding and provides a well-reasoned response to the oral defense question. The answer is clear, detailed, and demonstrates a good grasp of the topic. The student articulates their thoughts effectively, addressing the question thoroughly with only minor gaps.	The student demonstrates an exceptional understanding of the topic and provides a comprehensive, insightful response to the oral defense question. The answer is highly detailed, logically structured, and well-articulated, showing a deep understanding of the subject matter. The student confidently addresses the question, providing clear and thoughtful explanations.

		[0]	[0.1 – 4.13]	[4.14 – 4.9]	[4.91 – 5.32]	[5.33 – 6.02]	[6.03 - 7]
	CLO3 – Oral Defense Question [7 Marks]		The student's response to the oral defense question is minimal, vague, or off topic. The student shows little understanding of the topic, and the answer lacks depth, clarity, and relevance.	The student attempts to answer the oral defense question, but the response is incomplete or lacks depth. The answer shows some understanding of the topic, but it may be superficial, unclear, or not fully relevant. The student may struggle to articulate their thoughts clearly.	The student provides a basic, relevant response to the oral defense question. The answer demonstrates an adequate understanding of the topic, with a clear and logical explanation. While the response is generally well-articulated, it may lack depth, detail, or sophistication.	The student shows solid understanding and provides a well-reasoned response to the oral defense question. The answer is clear, detailed, and demonstrates a good grasp of the topic. The student articulates their thoughts effectively, addressing the question thoroughly with only minor gaps.	The student demonstrates an exceptional understanding of the topic and provides a comprehensive, insightful response to the oral defense question. The answer is highly detailed, logically structured, and well-articulated, showing a deep understanding of the subject matter. The student confidently addresses the question, providing clear and thoughtful explanations.

		[0]	[0.1 – 4.13]	[4.14 – 4.9]	[4.91 – 5.32]	[5.33 – 6.02]	[6.03 - 7]
	CLO4 – Oral Defense Question [7 Marks]		The student's response to the oral defense question is minimal, vague, or off topic. The student shows little understanding of the topic, and the answer lacks depth, clarity, and relevance.	The student attempts to answer the oral defense question, but the response is incomplete or lacks depth. The answer shows some understanding of the topic, but it may be superficial, unclear, or not fully relevant. The student may struggle to articulate their thoughts clearly.	The student provides a basic, relevant response to the oral defense question. The answer demonstrates an adequate understanding of the topic, with a clear and logical explanation. While the response is generally well-articulated, it may lack depth, detail, or sophistication.	The student shows solid understanding and provides a well-reasoned response to the oral defense question. The answer is clear, detailed, and demonstrates a good grasp of the topic. The student articulates their thoughts effectively, addressing the question thoroughly with only minor gaps.	The student demonstrates an exceptional understanding of the topic and provides a comprehensive, insightful response to the oral defense question. The answer is highly detailed, logically structured, and well-articulated, showing a deep understanding of the subject matter. The student confidently addresses the question, providing clear and thoughtful explanations.

		[0]	[0.1 – 8.26]	[8.27 – 9.7]	[9.8 – 10.6]	[10.7 – 12.04]	[12.05 - 14]
	CLO5 – Oral Defense Question [14 Marks]		The student's response to the oral defense question is minimal, vague, or off topic. The student shows little understanding of the topic, and the answer lacks depth, clarity, and relevance.	The student attempts to answer the oral defense question, but the response is incomplete or lacks depth. The answer shows some understanding of the topic, but it may be superficial, unclear, or not fully relevant. The student may struggle to articulate their thoughts clearly.	The student provides a basic, relevant response to the oral defense question. The answer demonstrates an adequate understanding of the topic, with a clear and logical explanation. While the response is generally well-articulated, it may lack depth, detail, or sophistication.	The student shows solid understanding and provides a well-reasoned response to the oral defense question. The answer is clear, detailed, and demonstrates a good grasp of the topic. The student articulates their thoughts effectively, addressing the question thoroughly with only minor gaps.	The student demonstrates an exceptional understanding of the topic and provides a comprehensive, insightful response to the oral defense question. The answer is highly detailed, logically structured, and well-articulated, showing a deep understanding of the subject matter. The student confidently addresses the question, providing clear and thoughtful explanations.